

Machine learning methods for continuous glucose monitoring data and their potential in cardiovascular risk assessment

PhD dissertation

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Abbreviations

AGP: ambulatory glucose profile

AID: Automated insulin delivery

AUROC: Area Under the Receiver Operating Characteristic Curve

Black: non-Hispanic African Americans or Black

CGM: continuous glucose monitoring

CV: coefficient of variation

CVD: cardiovascular disease

eGFR: estimated glomerular filtration rate

GMI: glucose management indicator

HbA1c: hemoglobin A1c

IGLU: Interpreting Glucose Data from Continuous Glucose Monitors

LMEM: linear mixed effect model

LOCF: last observation carried forward

LSTM: long short term memory model

MBG: mean glucose

NT-proBNP: N-terminal Pro-B-type Natriuretic Peptide

RMSE: root mean square error

SCORE2-Diabetes: Systematic Coronary Risk Evaluation 2-Diabetes

SMBG: self-monitoring blood glucose

T1D: type 1 diabetes

T1DX: T1D Exchange

T2D: type 2 diabetes

TAR: time above range

TBR: time below range

TIR: time in range

UMAP: Uniform Manifold Approximation and Projection

White: non-Hispanic White

Introduction 1

1.1 Diabetes

Diabetes is a metabolic disease characterized by high blood glucose levels known as hyperglycemia.^{1,2} The disease is one of the fastest growing global health challenges.³ In 2021 it was estimated that 537 million people lived with diabetes, and it is expected to reach 643 million by 2030.³ The most common forms of chronic diabetes are type 1 diabetes (T1D) and type 2 diabetes (T2D).¹ T1D is an autoimmune disease characterized by the destruction of the pancreatic islet β -cells, leading to an absolute insulin deficiency.⁴ Approximately 85%-95% of all people with diabetes in developing countries have T2D.¹ It is the most common form of diabetes and in contrast to T1D it generally begins with insulin resistance in which the body's cells become less responsive to insulin.^{1,5} Initially, the pancreas compensates by increasing insulin secretion, but over time progressive β -cell dysfunction results in inadequate insulin production to meet the body's demands.^{1,2,5} T2D is the most common form of diabetes. Despite their distinct underlying pathophysiology, both T1D and T2D are characterized by chronic hyperglycemia.

Diabetes can be diagnosed based on a bloodtest to check hemoglobin A1c (HbA1c) levels or through plasma glucose criteria (Table 1.1).^{4,6}

Although both plasma glucose tests and HbA1c tests are appropriate for diabetes screening, the HbA1c test has several advantages.⁴ Some of the benefits are that it requires no fasting and it is less sensitive to stress and changes in nutrition.⁴ HbA1c is an indirect measure of blood glucose levels because glucose binds hemoglobin over ~ 120 days lifespan of erythrocytes (red blood cells).⁴ However HbA1c testing is not always accessible, affordable or suitable (e.g., for people with anemia or under HIV-treatment).⁴ With the guidelines recommendation of using HbA1c, a debate has sparked on the controversy of using a set thresholds as it is well known that racial differences in HbA1c exists.⁷ Black individuals have higher HbA1c levels than Hispanic and non-Hispanic White people with diabetes,^{4,8} however studies show that the association between HbA1c and risk of complications appear to be similar between ethnic race groups.⁴

1.2 Blood Glucose Monitoring

HbA1c values are a very important part of diabetes management. The treatment goal is for HbA1c to be below 53 mmol/mol for non-pregnant adults without any hyperglycemia or hypoglycemia (low blood glucose levels). This corresponds to an estimated average glucose of 8.6mmol/L. However HbA1c goals should be individualized to fit each individual's specific situation and may be higher or lower than 53 mmol/mol.^{9,10} There are two tools available for self-assessment of glycemia, the self-monitoring blood glucose (SMBG) device measuring capillary blood with finger-sticks, and the continuous glucose monitoring (CGM) measuring interstitial fluid continuously.^{10,11} Both monitoring devices have shown to improve glycemic control in people with diabetes, however the CGM devices are superior when it comes to immediate information

Table 1.1: Diagnostic tests for diabetes

Test	Criteria	Description
Fasting plasma glucose	≥ 7.0 [mmol/L]	Fasting is defined as no calorie intake for at least eight hours.
2-hour plasma glucose	≥ 11.1 [mmol/L]	Glucose is measured 2 hours after oral ingestion of 75g glucose dissolved in water (OGTT)
Random plasma glucose	≥ 11.1 [mmol/L]	Individuals with classic symptoms of hyperglycemia (e.g., dehydration and unintentional weight loss) or hyperglycemic crisis (e.g., diabetic ketoacidosis) together with a plasma test taken any time of day.
Glycated hemoglobin (HbA1c)	≥ 48 [mmol/mol]	

and alerts on asymptomatic hypo- and hyperglycemic events and provide a comprehensive glucose profile including both pre- and postprandial blood glucose measurements.^{10,12,13}

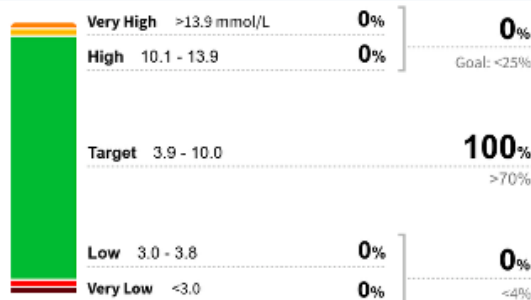
The CGM devices measures the glucose in the interstitial fluid, which is the fluid around and between cells. It corresponds with plasma glucose although a lag can appear when glucose levels rise or fall rapidly.¹⁴ Use of CGM devices has shown to improve glycemic control in both T1D and T2D with improvements in HbA1c levels and more time spent in target range (3.9-10 mmol/L) regardless of age, sex, education and income.^{9,14} CGM devices focus on monitoring and it is possible to get an ambulatory glucose profile (AGP) report from the devices (Figure 1.1). The AGP report shows summary statistics over all data (usually a two weeks period) including CGM-derived metrics as shown in (Table 1.2).¹² Besides monitoring, data from CGM devices are now used in combination with insulin delivery systems to automate the process of administering insulin.¹⁴ Automated insulin delivery (AID) systems typically use data from CGM devices to determine or adjust insulin delivery. The simplest AIDs are threshold-based low-glucose suspend systems in which insulin delivery is automatically suspended when blood glucose levels drop below a predefined threshold.¹⁵ More advanced systems utilize additional information such as carbohydrate intake and physical activity to optimize insulin delivery.¹⁷

Glucose Pattern Insights

Selected Dates: 11 Apr 2024 - 24 Apr 2024 (14 Days)

LibreView

Time in Ranges



Glucose Statistics

Average Glucose

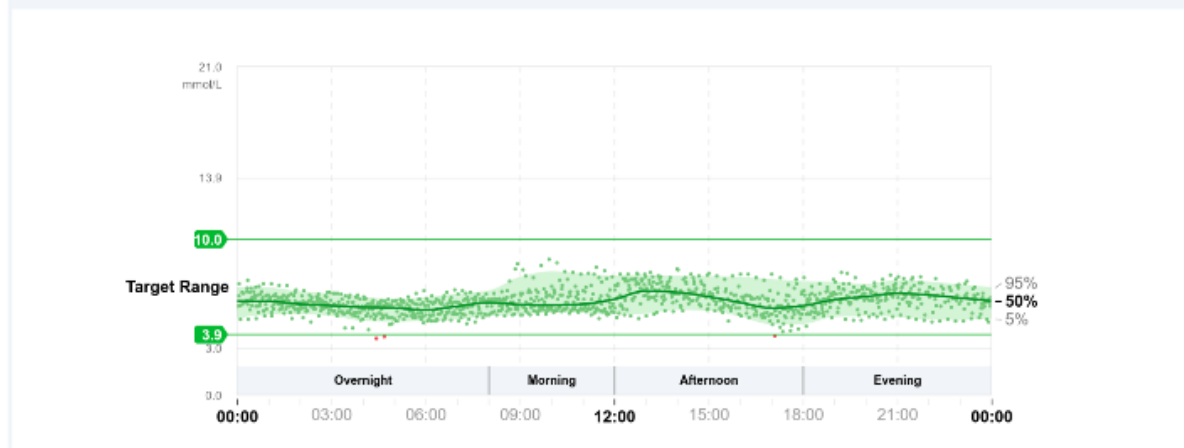
6.0 mmol/L Goal: ≤8.6 mmol/L

Glucose Management Indicator (GMI)

Approximate A1C level based on average CGM glucose level.

5.9% Goal: ≤7.0% | 41 mmol/mol Goal: ≤53 mmol/mol

Glucose Patterns (14 Days)



DAILY GLUCOSE PROFILES

Each daily profile represents a midnight to midnight period with the date displayed in the upper left corner.

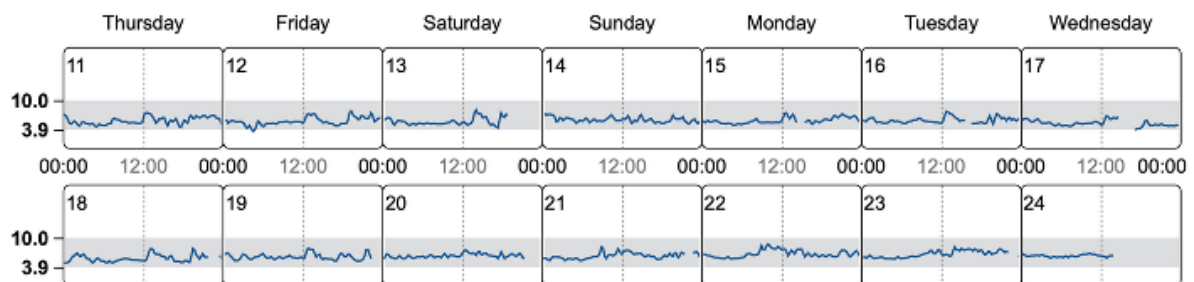


Figure 1.1: Ambulatory glucose profile report from female without diabetes. Data points are from the author herself using an intermittently-scanned (Manual data upload) FreeStyle LibreLink sensor.

1.3 Glycemic Variability

The challenge in obtaining good glycemic control is to keep blood glucose levels low enough to prevent comorbidities while avoiding the acute symptoms and, for insulin treated people with diabetes, potential mortality of iatrogenic (treatment-induced) hypoglycemia.¹⁸ The variability in the blood glucose fluctuations needs to be reduced in order to lower the average blood glucose levels while still avoiding hypoglycemia.¹⁸ Advances in CGM technology have enabled the quantification of glucose variability, a measure of the extent of fluctuations in blood glucose levels over time,

Table 1.2: CGM-derived metrics presented in the ambulatory glucose profile report with target values.

Abbreviation	CGM-derived metric	Range	Recommended target
GMI	Glucose Management Indicator		No specific target
TIR	Time in Range	3.9-10.0 mmol/L (70-180 mg/dL)	> 70%
TBR ₅₄	Time Below Range	< 3.0 mmol/L (< 54 mg/dL)	< 1%
TBR ₇₀	Time Below Range	< 3.9 mmol/L (< 70 mg/dL)	< 4%*
TAR ₁₈₀	Time Above Range	> 10.0 mmol/L (> 180 mg/dL)	< 25%**
TAR ₂₅₀	Time Above Range	> 13.9 mmol/L (> 250 mg/dL)	< 5%
CV	Coefficient of Variation		≤ 36%
	Number of days sensor was worn		≥ 14 days for pattern management
	Active sensor time		70% of 14 days

* This goal include both TBR₅₄ and TBR₇₀.

** This goal include both TAR₁₈₀ and TAR₂₅₀

including both peaks and nadirs.¹⁹ Describing the variability is clinically relevant because two individuals with similar HbA1c levels may exhibit different glycemic variability profiles.¹⁸ Numerous metrics have been developed to characterize different aspects of glucose variability, but no single measure captures all of its dimensions. Consequently, no universal consensus has been reached, although the coefficient of variation (CV) and SD have been incorporated into AGP reports and the American Diabetes Association guidelines (Table 1.2).^{18,19}

1.4 Cardiovascular Diseases and Heart Failure

Although advances in diabetes management have reduced the incidence of several diabetes-related complications, cardiovascular disease remains a major cause of morbidity and mortality in people with diabetes.^{20–22} Chronic hyperglycemia can lead to long-term complications, although the progression and manifestation of these complications vary between individuals.⁴ Identifying individuals at increased risk of atherosclerotic cardiovascular disease or heart failure before clinical disease develops is therefore an important component of diabetes care.²¹

Heart failure is at least twice as prevalent among people with diabetes as in the general population.^{21,23} It may present as either an acute or a chronic condition which is characterized by a combination of symptoms, including breathlessness, fatigue, and ankle swelling, together with clinical signs such as elevated jugular venous pressure and pulmonary crackles.²⁴ In individuals with suspected chronic heart failure based on risk factors, symptoms, clinical findings, and supporting investigations such as electrocardiography, measurement of N-terminal Pro-B-type Natriuretic Peptide (NT-proBNP) is recommended as part of the diagnostic work-up.²⁴ An NT-proBNP concentration of 125 pg/mL indicates the need for further cardiac evaluation, whereas concentrations below this threshold make chronic heart failure unlikely and alternative

diagnoses should be considered.²⁴ Consequently, NT-proBNP represents a clinically relevant biomarker for identifying individuals at increased risk of chronic heart failure.

1.5 Clinical Prediction Models

To facilitate early identification of individuals at increased cardiovascular risk, several clinical prediction models have been developed.^{22,25–27} By estimating the probability of a cardiovascular disease (CVD) event within a specified time horizon, these models can increase awareness of cardiovascular risk, encourage adherence to recommended lifestyle changes and medical treatment, and help identify individuals who are most likely to benefit from preventive interventions.²⁸ Among these, the Systematic Coronary Risk Evaluation 2-Diabetes (SCORE2-Diabetes) model estimates the 10-year risk of CVD in individuals with T2D. It was developed and calibrated using European cohorts, making it particularly relevant for the Danish cohort included in this dissertation.²² However, the SCORE2-Diabetes model relies solely on traditional clinical variables, including age, sex, smoking status, systolic blood pressure, high-density lipoprotein cholesterol, age at diabetes diagnosis, HbA1c, and estimated glomerular filtration rate (eGFR), and does not incorporate information on glucose dynamics beyond the long-term glycemic exposure reflected by HbA1c.²² HbA1c has been independently associated with risk of heart failure,²⁹ however it does not capture the short-term glucose fluctuations, which may be associated with the progression of vascular complications in diabetes.³⁰

1.6 Association vs Prediction

Association is not prediction and this distinction is important, as association and prediction are fundamentally different objectives, despite the terms sometimes being used interchangeably in the medical literature.³¹ Association studies aim to identify relationships between variables and an outcome at group level, whereas prediction models are developed to estimate future outcomes for individual patients.³¹ Traditional statistical modelling is often used to test hypotheses regarding associations between exposures X and outcomes Y .³² Regression models applied to observational data are commonly used to evaluate hypotheses derived from prior knowledge, where the choice of exposures is guided by existing theory.³² These approaches assume that the chosen statistical model adequately represents the underlying data-generating mechanism, allowing conclusions to be drawn from the estimated model parameters.³³ Results from association studies are typically summarized using estimated effect measures, such as regression coefficients, odds ratios, hazard ratios, or risk ratios, together with confidence intervals and significance tests,³¹

In contrast, prediction modelling applies statistical models to predict future outcomes Y based on a set of predictors X .³³ Prediction models may be implemented using a wide range of approaches, including regression models, decision trees, and machine learning algorithms.^{32,33} The prediction models are fitted to data (the predictors) and the output Y' is evaluated based on your outcome Y ,³³ and evaluation metrics on prediction studies often include classification metrics such as Area Under the Receiver Operating Characteristic Curve (AUROC), sensitivity, specificity and calibration plots³⁴ or correlation coefficients, and root mean square error (RMSE), for forecasting models.³⁵

1.7 Machine Learning and Representation Learning

Associations between CGM-derived metrics and CVD outcomes have been investigated in the literature,^{36–38} whereas evidence regarding the predictive value of CGM-derived metrics for cardiovascular risk assessment remains limited.^{39,Hua2012?,Koroleva2022?} The widespread adoption

of CGM has generated large volumes of glucose data, enabling the application of machine learning methods in diabetes care.^{16,40} Machine learning models are often divided into supervised and unsupervised machine learning.⁴¹ Supervised machine learning uses labelled data where each sample (e.g. age and sex from one individual) has a label or an outcome (e.g., elevated NT-proBNP) to predict new outcomes for previously unseen data. Unsupervised machine learning on the other hand uses data without labels to explore underlying structures or clusters in data.⁴¹ Among supervised learning methods, deep neural networks are particularly notable because they can automatically learn informative representations directly from raw data for downstream tasks such as prediction or classification, reducing the need for manual feature engineering. This process is known as representation learning.^{16,42} Consequently, deep neural networks offer an alternative to relying on predefined CGM-derived metrics. Although international consensus recommendations advocate the routine reporting of glucose variability using CV, MAGE, and SD, with particular emphasis on CV,¹² glucose variability encompasses multiple dimensions, including within- and between-day variability, the amplitude of glucose excursions, and temporal glucose patterns.¹⁸ As a result, no single metric has been shown to comprehensively capture all aspects of glucose variability, and no consensus currently exists regarding the optimal measure.^{18,19}

1.8 Foundation Models and Transfer Learning

With the scale and scope that some of the machine learning models have developed over recent years, a new term has arisen, foundation models.⁴³ These are defined as “*any model that is trained on broad data (generally using self-supervision at scale) that can be adapted (e.g., fine-tuned) to a wide range of downstream tasks*”.⁴³ Two CGM specialized foundation models exist in the literature which both use self-supervised machine learning.^{44,45} Self-supervised machine learning is a technique used to train models on unlabelled data through different techniques, one of them including masked learning (e.g. predicting manually masked words in text or values in time-series signals).^{43–45} A foundation model is first pretrained on a general task, such as predicting masked values in CGM data, allowing it to learn representations of glucose dynamics. These representations can subsequently be adapted to a downstream task, such as predicting cardiovascular outcomes, through transfer learning.^{43,45} A foundation model can be conceptualized as comprising a pretrained backbone and a task-specific head (Figure 1.2).⁴⁶ The backbone extracts representations from the input data, whereas the head uses the representations to perform the downstream task. Transfer-learning strategies differ primarily in how the pretrained backbone is used. It may remain fixed and serve as a feature extractor, with only a downstream model or task-specific head being trained, or some or all of the model may be updated through fine-tuning.^{46,47} The choice of strategy depends partly on the similarity between the pretraining and downstream tasks.⁴⁷ One of the main challenges of deep neural networks is their requirement for large amounts of training data. Transfer learning addresses this challenge by reusing knowledge learned from large datasets and applying it to related tasks with limited labelled data. This is particularly relevant in clinical risk prediction, where labelled data are often scarce, especially for minority populations.^{46,48,49}

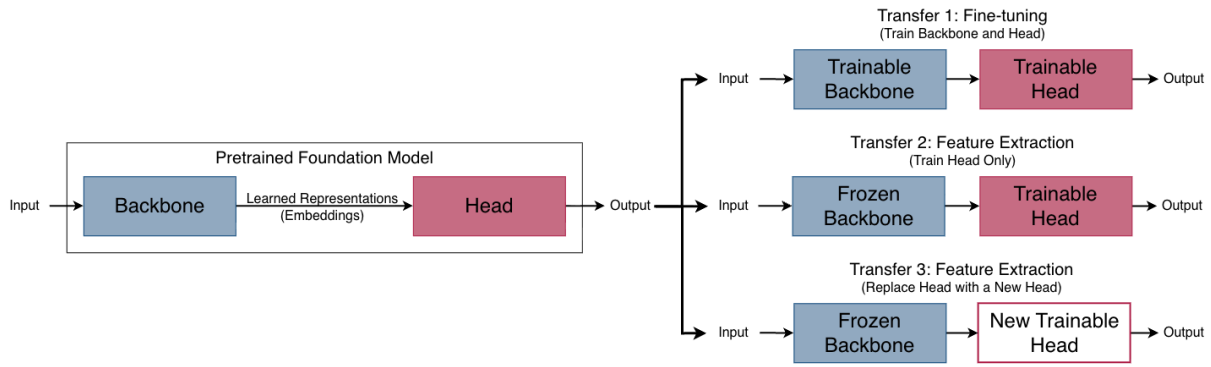


Figure 1.2: Overview of transfer learning strategies. Inspired by Deng et al.⁴⁶ under the Creative Commons Attribution 4.0 International (CC BY 4.0) licence.

1.9 Health Disparities and Bias in Machine Learning

Transfer learning becomes particularly relevant in clinical risk prediction, where labelled data are often scarce, especially for minority populations.^{46,48–50} A study by Turner et al.⁵⁰ showed that among 20,692 US-based clinical trials, only 43% reported participants' race or ethnicity. Although the reporting of race and ethnicity increased between 2010 and 2020, racial and ethnic minority groups remained underrepresented relative to their proportions in the US population.⁵⁰ Furthermore, this lack of information is not only present in clinical trials but also in reportings of FDA-approved AI medical devices.^{49,50} A study by Muralidharan et al.⁴⁹ found that only 3.6% of the FDA approval documents (Summary of Safety and Effectiveness Data Documents) provided information on race and/or ethnicity of the intended or testet users.⁴⁹ This is problematic in data-driven models such as machine learning models since they reflect the populations represented in their training data and when minority groups are underrepresented, the resulting models may not generalize equally well to all patients.^{49,51}

This challenge is particularly relevant in diabetes research. Although diabetes affects individuals across all racial and ethnic groups, differences have been reported in glycemic response, glycemic control, insulin resistance and the burden of diabetes-related complications between populations.⁷ Consequently, models developed predominantly using one population may not achieve comparable predictive performance in others and potentially reinforcing existing healthcare disparities.⁵¹ For example, risk prediction models have been shown to underestimate the risk of type 2 diabetes among non-Hispanic Black individuals,⁵¹ and algorithmic bias has been demonstrated to substantially influence the allocation of healthcare resources.⁵⁵ Developing equitable prediction models therefore requires both representative data and careful evaluation of model performance across demographic groups.

2 Aims

The overall aim of this PhD dissertation was to investigate the potential of continuous glucose monitoring (CGM) data to improve cardiovascular risk assessment and to evaluate the role of machine learning in extracting clinically relevant information from CGM data across different diabetes populations, including the evaluation of racial disparities in model performance.

2.1 Specific Aims

Aim 1: To identify and summarize the existing literature on associations between glycemic control and cardiovascular outcomes, including the use of CGM-derived metrics for cardiovascular risk prediction. (*Study I*)

Aim 2: To determine whether a CGM foundation model learns meaningful glucose representations in individuals with type 1 and type 2 diabetes. (*Study II*)

Aim 3: To evaluate whether traditional CGM-derived metrics and learned glucose representations improve prediction of elevated NT-proBNP beyond established clinical risk factors in individuals with type 2 diabetes. (*Study II*)

Aim 4: To evaluate the fairness of machine learning models for glucose prediction by assessing model performance across non-Hispanic Black and non-Hispanic White populations. (*Study III*)

In this chapter the three studies included in this PhD will be presented together with a description of the datasets and methods used in the studies.

3.1 Overview of Studies

This dissertation includes three studies which are summarized in Table 3.1.

Table 3.1: Overview of the included studies

	Study I	Study II	Study III
Title	Continuous Glucose Monitoring-Derived Metrics and Cardiovascular Risk Among People With Diabetes: Systematic Scoping Review	Foundation model-derived representations of continuous glucose monitoring data: relation to glycemic patterns and cardiovascular risk	Racial disparities in continuous glucose monitoring-based 60-min glucose predictions among people with type 1 diabetes
Study Type	Review	CGM foundation model evaluation study	Algorithmic fairness study
Data Sources	Literature on Embase and MEDLINE	The CANCAN study and the T1D Exchange (T1DX) dataset	T1DX dataset
Population	Studies including people with type 1 diabetes, type 2 diabetes and/or prediabetes	159 people with type 2 diabetes 205 people with type 1 Diabetes	205 people with type 1 Diabetes
Input, predictors and exposures	Search strategy and eligibility criteria	Clinical variables, CGM-derived metrics and CGM data	CGM data
Outcome, target	All studies that uses CGM data to investigate cardiovascular disease risks and outcomes	Visualization of foundation model embeddings and elevated NT-proBNP prediction	60-minute glucose prediction

	Study I	Study II	Study III
Analysis	Systematic scoping review, study characterization, and narrative synthesis	Foundation models and logistic regression models	long short term memory model (LSTM) and linear mixed effect model (LMEM)

3.2 Scoping Review Methodology

A scoping review was conducted to investigate current knowledge of CGM-data and its possible connection and predictive power to clinical and subclinical cardiovascular outcomes. Given the expected heterogeneity between studies and the broad exploratory nature of the research question, a scoping review was considered the most appropriate methodology, as the aim was to map the existing evidence within this field.⁵⁶ The scoping review followed the Manual for Evidence Synthesis (Chapter 10 - Scoping reviews) from the Joanna Briggs Institute (JBI) and reported according to the Preferred Reporting Items for Systematic Reviews and Meta-analyses extension for scoping review (PRISMA-ScR) guidelines.^{56,57}

3.2.1 Concepts and Definitions

The review focused exclusively on studies using CGM data as input for analyses of associations with, or prediction of, CVD. This meant that studies only investigating self-monitored blood glucose with finger-prick measurements or blood sample measurements such as HbA1c were not considered in this review. CVD outcomes were divided into subclinical or clinical. For the clinical CVD, the following outcomes were considered: cardiovascular mortality, major adverse cardiovascular events, coronary artery disease, heart failure, stroke, and peripheral artery disease with synonymous terms included (e.g., ischemic heart diseases) and also clinical events (e.g., undergoing coronary artery bypass surgery). Subclinical CVD was grouped into six subcategories: arterial stiffness, flow resistance, arterial wall thickness, arterial wall composition, cardiac and pulse-related measures, and arterial lumen. CVD outcomes did not include broader risk factors nonspecific to cardiovascular risk (e.g., age or sex). Eligibility criterias for the scoping review³⁷ (Study I) can be found in Table 3.2.

3.2.2 Search and Selection of Sources

MEDLINE and Embase were searched from inception to March 11, 2025. The two databases were selected because they are core databases for biomedical literature. Owing to the resources available to the review team, the search could not be extended to additional databases. The search strategy was developed in collaboration with information specialists. An exploratory search was held and 13 studies within the field was identified and used to test the search strategy by ensuring the 13 identified studies were included in the final search string.^{36,39,58–68} The search strategy was wished to be kept broad as it was known beforehand that the field contained inconsistent terminology and overlapping search categories (e.g., ‘blood glucose monitoring’ and ‘glycemic control’ both encompass finger-prick measurements) in order to ensure identification of all the relevant studies in MEDLINE and Embase. The full search strategy can be found in Appendix A.

All identified records were uploaded into EPPI reviewer⁶⁹ and duplicates were removed. Titles and abstracts were screened by at least two independent reviewers and a meeting was held after ~5% of the titles and abstracts had been screened, to resolve disagreements and ensure consistency between reviewers. Afterwards full-text screening was performed by two independent

Table 3.2: Eligibility criteria for the scoping review (Study I)

Inclusion Criteria	Exclusion Criteria
Human clinical studies including participants with prediabetes or any type of diabetes except gestational diabetes, regardless of definitions. Peer-reviewed published original articles (including brief reports). Studies investigating either: a) The association between CGM-derived metrics of glycemic control and cardiovascular risk markers or cardiovascular diseases (prevalent or incident). b) CGM-derived metrics of glycemic control as predictors of cardiovascular risk markers or cardiovascular diseases (prevalent or incident).	Review articles, editorials, case reports, protocols, conference abstracts and preprints. Animal studies not including human participants. Studies not including metrics derived from CGM device data. Studies including CGM-derived metrics as outcomes. Studies not focusing on CVD outcomes according to the definition in <i>Concepts and Definitions</i> section. Studies focusing on pregnant women with any form of diabetes, including gestational diabetes. Studies where participants were monitored post-surgery or during hospitalization (e.g., intensive care unit). Language not understood by the authors.

reviewers and held up against eligibility criteria. Disagreements were resolved through discussion at physical meetings between reviewers.

Backward and forward citation searching was performed on all the included studies after the screening process using citationchaser⁷⁰ and the screen process was repeated until no further new studies were identified through citationchaser.

3.2.3 Data Charting Process and Data Items

Research questions were defined before the screening process and data was extracted based on the research questions in Note 1.

Note 1: Research Questions (Study 1)

1. Is there an association between glycemic control and CVD risk?
2. Can CGM-derived metrics predict CVD risk?
3. What CGM-derived metrics are used in the literature?
4. Which cardiovascular markers and diseases are included as outcomes in the studies?
5. What characterizes study populations (age, sex, ethnicity, geographic location)?
6. What study designs are used (e.g., longitudinal cohort studies, randomized controlled trials, cross-sectional studies)?
7. How was data collected (e.g., clinical trial, epidemiological study, routinely collected data)?
8. What CGM devices were used?
9. What statistical models were used in the studies?
10. Is the data openly available?
11. Is the code openly available?

3.2.4 Synthesis of Results

Study characteristics were synthesized using descriptive statistics and supplemented by a narrative summary. Only the most adjusted models in the main population for each study was considered in this review. This approach was necessary as several studies reported numerous estimates across varying adjustments, different combinations of CGM-derived metrics and CVD outcomes, and/or distinct subgroup stratifications. Given the large volume of literature, studies investigating clinical cardiovascular outcomes were prioritized in the main synthesis. While subclinical markers offer valuable insight into early vascular changes and disease progression, clinical outcomes are more directly relevant to cardiovascular risk prediction and clinical decision-making. The main synthesis of clinical outcomes included both effect estimates and measures of statistical significance whereas the supplementary synthesis on subclinical CVD focused primarily on creating an overview of the chosen CGM-derived metrics and the statistical significance of reported associations. All evaluation metrics mentioned in the studies doing prediction analyses were reported based on what the studies reported.

3.3 Study Populations

- ☐ Signe, do you have ethical approval number or paper/protocol I can cite for the CANCAN study?

Two datasets were used in this PhD project and are described in the following sections. Both studies were conducted in accordance with the principles of the Declaration of Helsinki and approved by the relevant institutional review boards. Written informed consent was obtained from all participants.

3.3.1 CANCAN Data

The CANCAN dataset was collected in an observational case-control study from three hospital outpatient clinics (Denmark). The aim of the CANCAN Study was to investigate if cardiac autonomic neuropathy (CAN) assessments, CGM and heart failure indicators could be used in risk stratification in people with T2D. Participants were included if they were adults (> 18 years old) and had T2D for at least a year. Exclusion criteria were atrial fibrillation or laser-treated eye disease within the past three months, were pregnant or lactating, had a life-threatening disease (expected survival < 1 year), or had cognitive disabilities rendering them unable to understand and give consent. Data were collected during a clinical examination, during which blood and urine samples were obtained, and a blinded CGM sensor was inserted. Blood samples to obtain NT-proBNP values were collected when the CGM sensors were returned, after the 2-week CGM data collection period had ended (Figure 3.1). In total, 159 participants were included since the rest were missing age, sex, HbA1c, BMI, NT-proBNP, HDL, total cholesterol, or eGFR and these values were needed for the analyses in Study II.

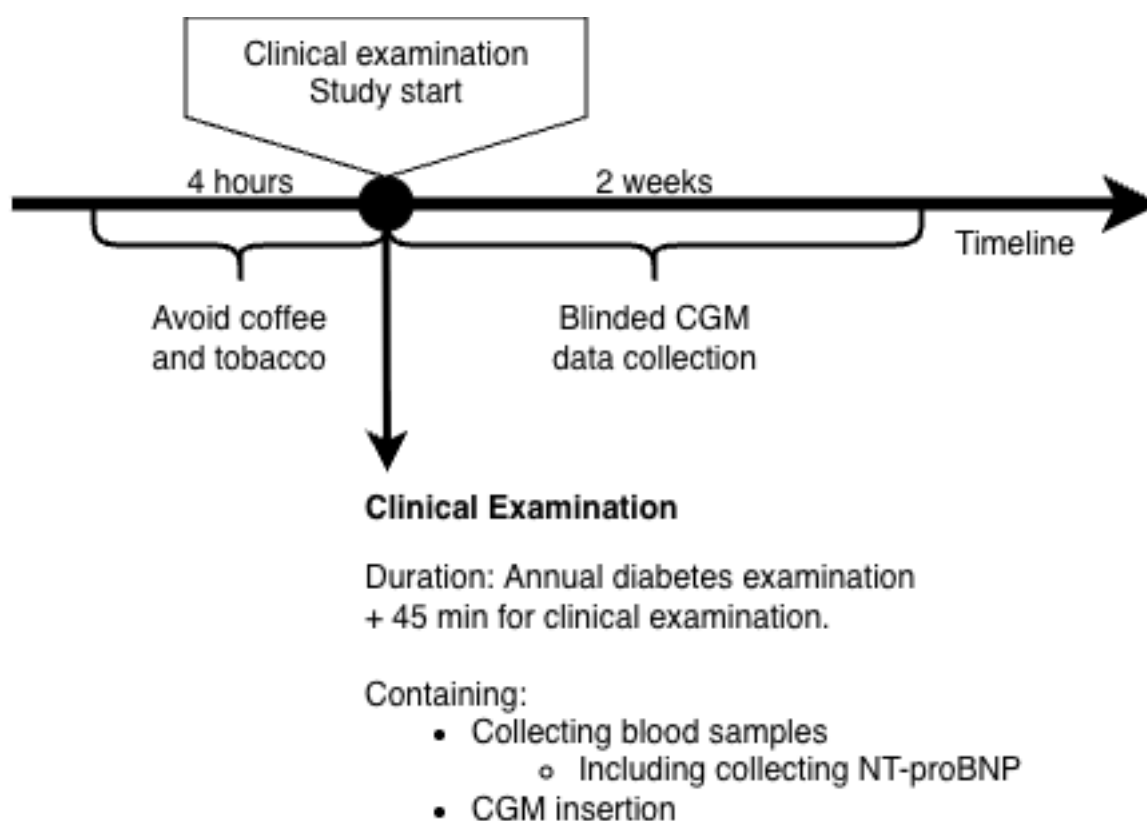


Figure 3.1: Data collection timeline for the CANCAN Study

3.3.2 T1D Exchange Data

The publicly available T1D Exchange (T1DX) dataset, distributed by the Jaeb Center for Health Research under the protocol “Racial Differences in Mean CGM Glucose in Relation to HbA1c, was collected in 10 diabetes centers (US).⁷¹ The data was collected to determine if there is a racial difference in the association between mean glucose and HbA1c between non-Hispanic African Americans or Black (Black) and non-Hispanic White (White) people with T1D. In the T1DX study ethnicity and race was defined through self-identified and self-reported ethnicity/race verification forms where ethnicity was defined as Hispanic or Latino as: “a person of Cuban, Mexican, Puerto Rican, South or Central American, or other Spanish culture or origin, regardless of race. The term “Spanish origin” could be used in addition to “Hispanic” or “Latino” “. ^{54, suppl} For race a participant defined as White was “a person having origins in any of the original peoples of Europe, the Middle East or North Africa.” and Black or African American people were defined as “a person having origins in any of the Black racial groups of Africa. Terms such as “Haitian” or “Negro” can be used in addition to “Black” or “African American” “. ^{54, suppl} In this thesis, the term race will be used moving forward, as it aligns with the terminology used by Bergenstal et al.⁵⁴. However, I acknowledge that it is difficult to define race.⁷² Of the 205 participants included in the study⁵⁴, 101 were White and 104 were Black. For each race, two age groups were sampled 8 to <18 years old and >18. Each participant wore a blinded modified investigational version of Abbott Diabetes Care’s Flash Glucose Monitoring System (CGM) as continuously as possible for 12 weeks (Figure 3.2). CGM data was sampled every 15 minutes in a range between 2.21 - 27.65 mmol/L (40 and 498 mg/dL). Blood samples were taken at enrollment, at some of the follow-up visits, and at the end of enrollment.

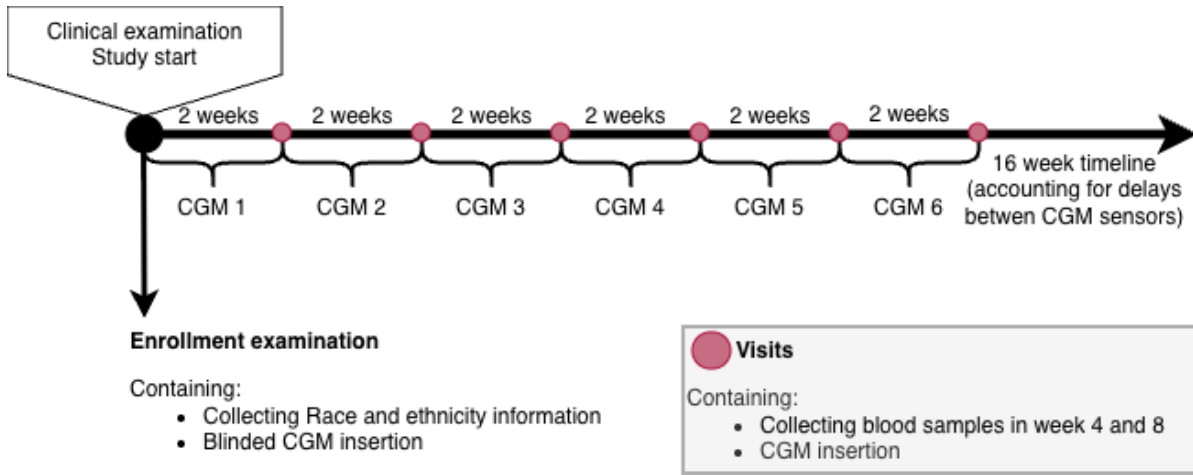


Figure 3.2: Data collection timeline for the T1DX Study

3.4 Foundation Models and Predictive Utility of CGM data

The second study investigated whether a CGM foundation model could learn meaningful representations of CGM data and compared these learned representations with conventional CGM-derived metrics identified in Study I.³⁷

3.4.1 Data and Participant Selection

- Check cancan data to find the range of the CGM censor in that study

Both the CANCAN and the T1DX dataset was used for Study II. Participants with missing age, sex, HbA1c or BMI were excluded from both cohorts and in addition, CANCAN participants with missing NT-proBNP, HDL, total cholesterol, and eGFR they were also removed, since it was necessary to calculate the SCORE2-Diabetes risk score. T1DX participants with missing race data was also removed. Days with less than 70% of CGM data was removed as well as participants with less than one day of data from both datasets to ensure sufficient data for the analyses. Values exceeding the measurable range of the CGM devices were retained as the device's upper and lower measurement limits. *For the CANCAN data, this range was (_____) and for the T1DX data, the range was 2.21 - 27.65 mmol/L.*

3.4.2 Transfer Learning, Foundation Models and Embeddings

Of the two CGM-specific foundation models based on self-supervised learning currently described in the literature (CGMformer by Lu et al., 2025⁴⁴ and GLUformer by Lutsker et al., 2026⁴⁵), only the pretrained weights from the CGMformer by Lu et al.⁴⁴ were made publicly available, enabling reuse of the CGMformer model for downstream tasks. The CGMformer was therefore chosen and the data was organized in 24-hour long-format input to match the input requirements of the CGMformer. The output of the foundation model is learned representations of the input data, also known as embeddings and the head of the CGMformer were not available for retraining (Figure 3.3). By only having the backbone, the transfer learning strategy being used had to be replacing the head with a new, as shown in Figure 1.2 as transfer three.

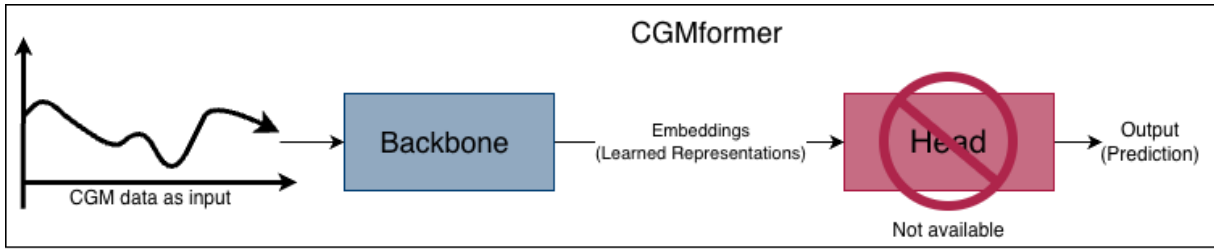


Figure 3.3: Illustration of how CGMformer is used in this analysis

Embeddings are numerical vectors that summarize the characteristics of the input glucose profile. A simple vector can be drawn as an arrow in a 2-dimensional space as shown in Figure 3.4, however the CGMformer gives a 128-dimensional embedding vector for each 24-hour CGM data input which is not easily visualized.

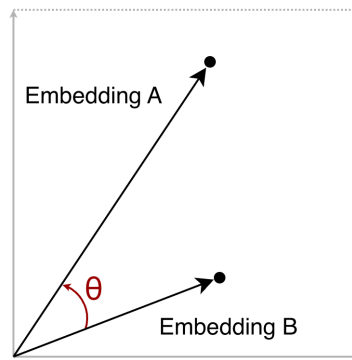


Figure 3.4: Embeddings shown as 2-dimensional vectors with the angle θ between the two embeddings.

Uniform Manifold Approximation and Projection (UMAP) algorithm was used to reduce the dimensionality of the embeddings to two dimensions. The concept of dimensionality reduction is illustrated in Figure 3.5. The UMAPs were visualized and compared with current literature, and later on used for prediction purposes.^{44,45,73} The UMAPs were colored by TIR, and MAGE to investigate if the embeddings were able to capture relevant information about glycemic patterns. History of CVD and elevated NT-proBNP for the CANCAN embeddings were also used to color the UMAPs to investigate if the embeddings were able to capture relevant information about cardiovascular risk and race was used to color the T1DX embeddings to investigate if the embeddings were able to capture relevant information about racial differences in glycemic patterns.

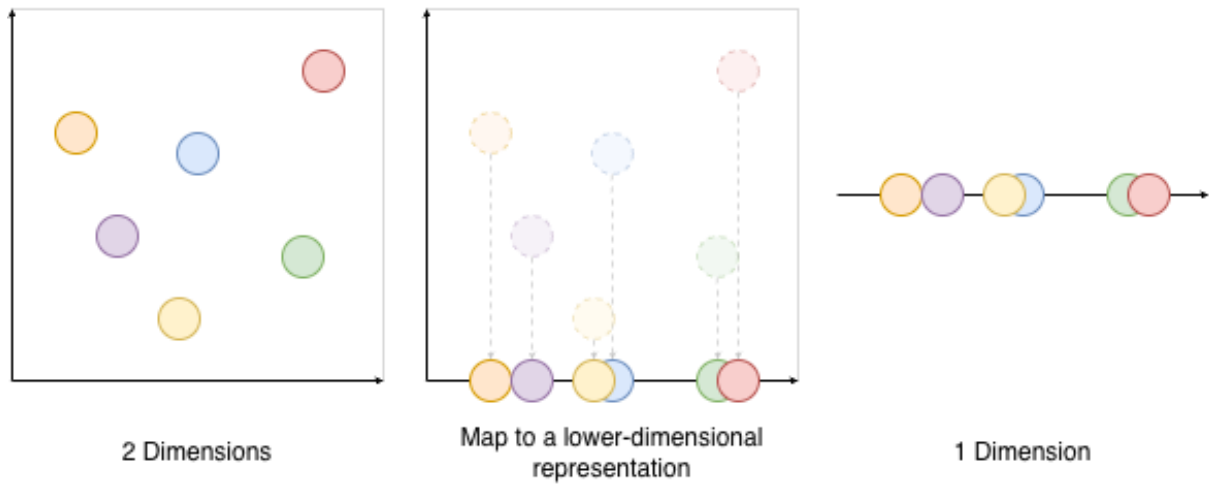


Figure 3.5: Conceptual illustration of dimensionality reduction (read from left to right). Information from multiple dimensions can be represented using fewer dimensions while preserving information. This illustration does not show how UMAP works but instead is intended to convey the general concept of dimensionality reduction.

Because embeddings are numerical vectors, they can be compared to quantify the similarity between glucose profiles, as demonstrated by Lutsker et al.⁴⁵ We therefore adopted a similar approach to assess the similarity between CGM-derived embeddings. Cosine similarity was used and it measures the angle between two embedding vectors independent of their magnitude (the length of the vectors). The cosine similarity is calculated by measuring the cosine angle between two vectors as shown by θ in Figure 3.4. The cosine similarity is a number between -1 and 1, where 1 means the two vectors are identical, 0 means they are unrelated, and -1 means that they are pointing in opposite directions (Figure 3.6). For the analyses in this study, cosine distance ($1 - \text{cosine similarity}$) was used because it expresses similarity as a distance measure, allowing embeddings with similar orientations to have small distances and facilitating downstream analyses and visualizations. The cosine distances range from 0 to 2 where 0 means the two vectors are pointing in the same direction, 1 means they are unrelated, and 2 means that they are pointing in opposite directions.

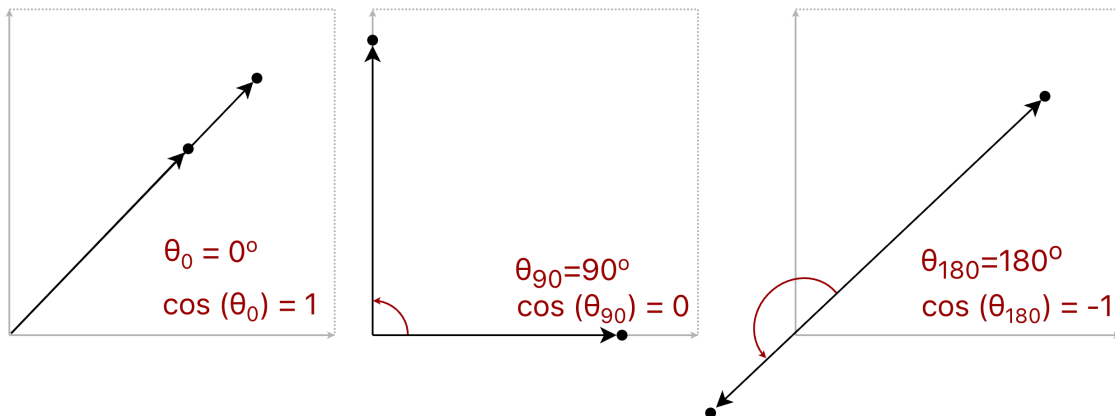


Figure 3.6: Embeddings shown with different cosine similarities. The cosine similarity is calculated as the cosine of the angle (θ) between two embeddings.

Kernel density plots were used to visualize the distribution of cosine distances within specific clinical strata and across differing participants (Figure 3.7). Groupings were made based on days

from the same participant (within-participant comparison), days from different participants with the same clinical strata (between-participant comparison), and days from different participants with different clinical strata (between-participant comparison) (Figure 3.7).

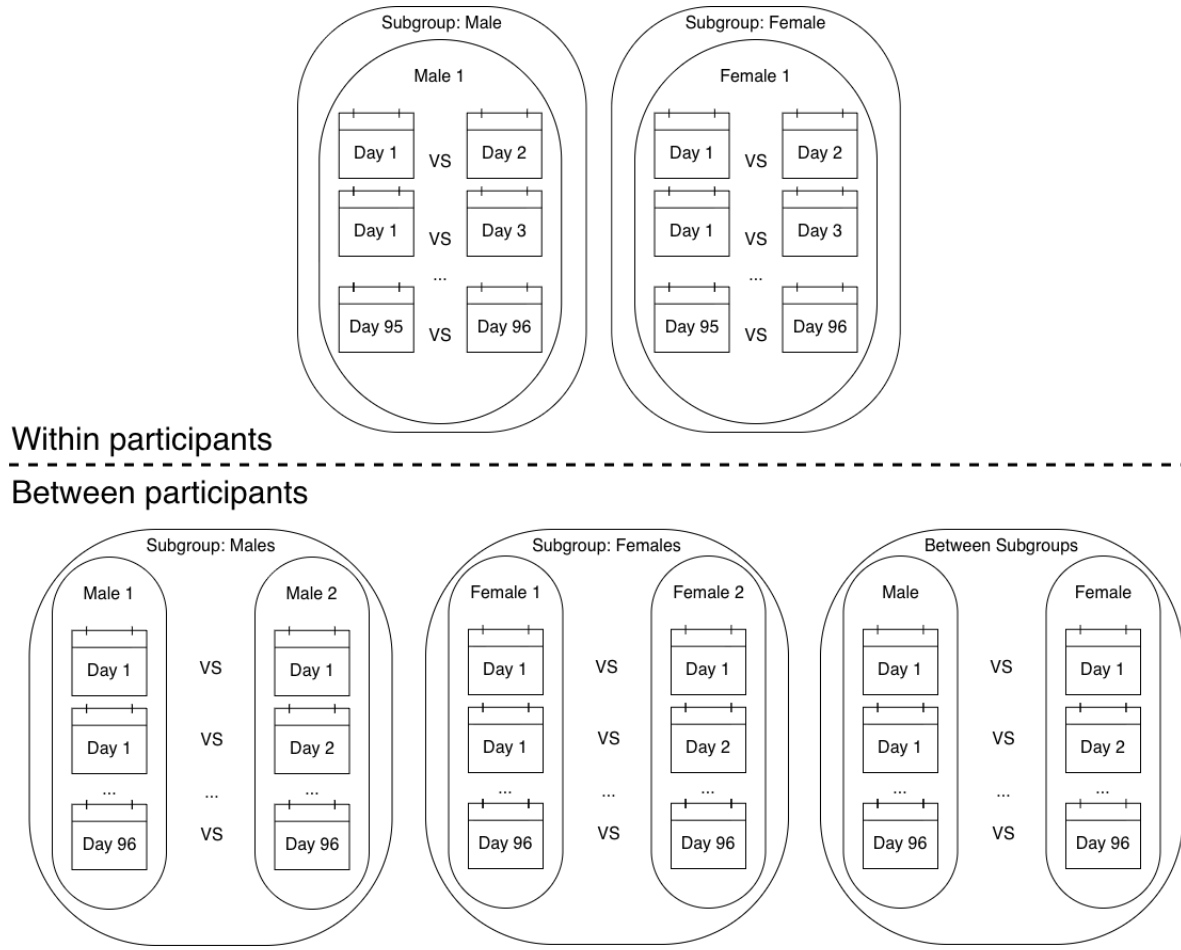


Figure 3.7: The sex strata is used as an example to illustrate the cosine distance analysis of the within and between cosine distance pairs

The main focus for the participant comparisons were TIR and HbA1c. The thresholds were as followed: TIR 70% (70–180 mg/dL, 3.9–10.0 mmol/L) and HbA1c <7.0 % (< 53 mmol/mol).¹⁴ Additional UMAP visualizations were generated by coloring participants according to race, MAGE (< 70 mg/dL (< 3.9 mmol/L)),¹² and NT-proBNP (NT-proBNP 125 pg/mL).⁷⁵ These categorizations were used solely to facilitate interpretation of the UMAP visualizations and were not incorporated into model development or any statistical analyses. Participants with missing values for a given characteristic were excluded only from the corresponding visualization and remained eligible for all subsequent analyses.

3.5 Cardiovascular Risk Assessment

The following section is about cardiovascular risk assessment evaluated predicting NT-proBNP as a binary outcome using a >125 pg/mL threshold.^{24,74,75} Only the cross-sectional CANCAN dataset was used for the cardiovascular risk assessment. The focus was to test the predictive power of CGM data both as represented as embeddings and as well-established CGM-derived

metrics. Some models also included clinical predictors to compare up against baseline models. Prediction models integrated clinical, standard CGM-derived, and learned representations.

3.5.1 Predictors

Inspiration was drawn from the SCORE2-Diabetes risk score, which is a 10-year CVD risk score for european people with diabetes. It was decided to use the input predictors of the SCORE2-Diabetes risk score as the clinical set of predictors used in this study.²² This resulted in set of clinical predictors including age, sex, smoking status, systolic blood pressure, total cholesterol, and HbA1c.²² Initial testing showed that the numbers of predictors had to be limited due to the small sample size and the SCORE2-Diabetes risk score was chosen as another way to represent the clinical predictor set while keeping the number of predictors down. Additionally age was tested as a predictor on its own to see if it was a strong predictor of elevated NT-proBNP, since NT-proBNP is known to increase with age.⁷⁶ Time in range (TIR) and mean amplitude of glycemic excursions (MAGE) were chosen as they were the CGM-derived metrics most commonly used in the literature based on the findings from Study I.³⁷ All CGM-derived metrics in this thesis were calculated using the Interpreting Glucose Data from Continuous Glucose Monitors (IGLU) package^{77,78} in R (default settings). TIR was calculated as the percentage of time spent in the range of 3.9–10.0 mmol/L (70–180 mg/dL).¹² MAGE was calculated as the mean amplitude of glucose excursions, where the included excursions were consecutive peak-to-nadir (or nadir-to-peak) glucose differences exceeding one standard deviation of the individual's glucose measurements. The IGLU implementation computationally reproduces the original manual procedure proposed by Service et al.⁷⁹ The CGMformer generated one 128-dimensional embedding for each 24-hour CGM recording. Because NT-proBNP was measured at the participant level, a single embedding was required for each participant. Therefore, embeddings from multiple recording days were aggregated using pooling to obtain one participant-level embedding. Both minimum, maximum, and mean pooling were explored to determine the most effective aggregation method. Maximum pooling was considered the default based on previous work.⁴⁵ The resulting 128-dimensional embeddings were subsequently reduced to two dimensions using UMAP, thereby enabling their use in the prediction models while keeping the number of predictors down to a minimum.

3.5.2 Model Development and Evaluation

Prediction models were developed using logistic regression to predict elevated NT-proBNP (>125 pg/mL) as a binary outcome. The models were evaluated using the area under the receiver operating characteristic curve (AUROC) as the primary metric. Bootstrap with optimism correction was employed for internal validation because of the limited sample size and the absence of an external validation cohort.^{80–82} The method repeatedly sampled bootstrap samples by sampling with replacement from the original dataset. For each bootstrap sample, the prediction model is fitted and its performance is evaluated both in the bootstrap sample (apparent performance) and in the original dataset (test performance). The difference between these two performance estimates represents the optimism due to overfitting. Averaging the optimism across all bootstrap samples and subtracting it from the apparent performance of the model fitted to the full dataset yields an optimism-corrected estimate of model performance. Internal validation was performed using 500 bootstrap resamples.^{80,82} Prediction stability plots were reported for each model to evaluate the stability of predictions produced by a specific model strategy when the training data changes.⁸²

3.5.3 Prediction Scheme Overview

The prediction models were structured hierarchically based on clinical variables (Model A) and the calculated SCORE2-Diabetes score (Model B) in the configurations shown in Table 3.3

Table 3.3: Overview of models and predictors

Model family	Model	Baseline predictors	Additional predictors
Clinical	A0	Age	–
Clinical	A1	Clinical variables*	–
Clinical	A2	Clinical variables*	TIR + MAGE
Clinical	A3	Clinical variables*	UMAP dimensions
SCORE2-Diabetes	B1	SCORE2-Diabetes score	–
SCORE2-Diabetes	B2	SCORE2-Diabetes score	TIR + MAGE
SCORE2-Diabetes	B3	SCORE2-Diabetes score	UMAP dimensions
CGM only	C1	TIR + MAGE	–
Embeddings only	D1	UMAP dimensions	–

* *Clinical variables are the same as those included in the SCORE2-Diabetes risk model: Age, sex, smoking status, systolic blood pressure, HDL cholesterol, total cholesterol, diabetes duration, HbA1c, and eGFR.*²²

TIR = Time in Range; MAGE = Mean Amplitude of Glycemic Excursions.

3.6 Algorithmic Fairness and Blood Glucose Prediction

The third study was investigating racial bias in a data driven machine learning model only based on blood glucose measurements from a CGM device.

3.6.1 Preprocessing of CGM data for Forecasting

A prediction horizon of 60 minutes was chosen because it provides sufficient time for preventive interventions for hypoglycemia and hyperglycemia, such as insulin dose adjustments or carbohydrate intake, while accounting for the 15-minute sampling interval of the CGM device and the physiological delay between blood and interstitial glucose measurements.¹⁴

Accordingly, the CGM data were divided into consecutive 60-minute windows containing four glucose measurements to predict glucose levels 60 minutes into the future, as illustrated in Figure 3.8. Prediction windows containing missing CGM values were excluded.

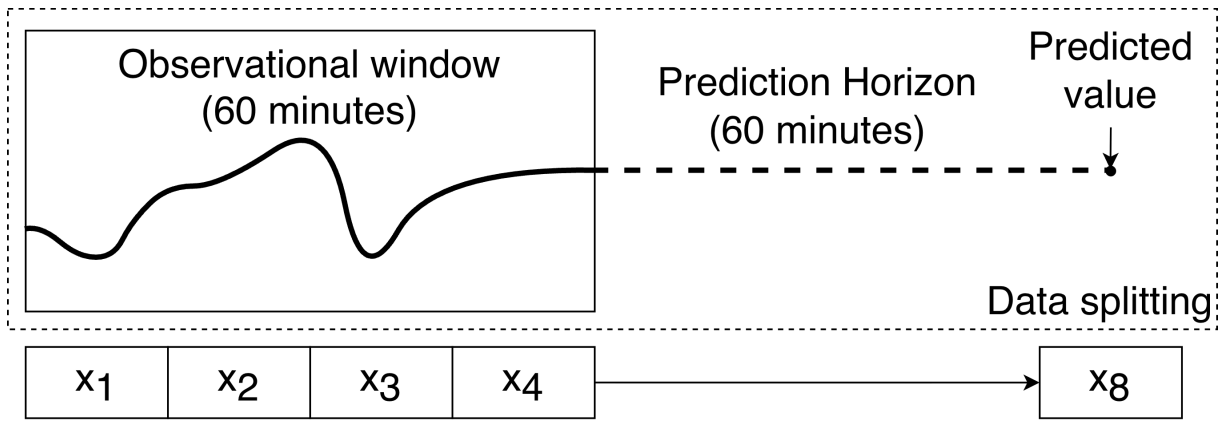


Figure 3.8: The prediction horizon and overview of the creation of samples (predictors and outcomes).

3.6.2 Partitioning of the Dataset

The T1DX dataset was randomly resampled without replacement at the participant level to avoid data leakage and to evaluate model performance on independent participants. As illustrated in Figure 3.9, training subsets with varying proportions of Black and White participants were generated by randomly selecting participants from the original dataset while keeping the train-test split at the participant level. Eleven training set compositions were created, ranging from 100% White/0% Black to 0% White/100% Black in 10% increments.

Because some training subsets consisted entirely of participants from a single racial group, the maximum training set size on participant level was constrained by the size of the smaller racial group minus the participants kept for testing. Consequently, each training subset contained approximately half the number of participants in the original T1DX dataset. Repeating this procedure for each participant resulted in 2,255 training datasets (205 participants $\times$ 11 training set compositions) and 205 independent test sets. Throughout all resampling procedures, all CGM recordings from a participant were kept together to prevent data leakage between the training and test sets.



Figure 3.9: The prediction horizon and overview of the creation of subsets for training.

Eleven subsets were created for each of the participants and the last week's data was chosen to test on as illustrated in Figure 3.10. To preserve sample size, one week was chosen because participants had varying amount of CGM data and the participant with the least amount of data had ~ 2 weeks. The remaining data was used for training the models.

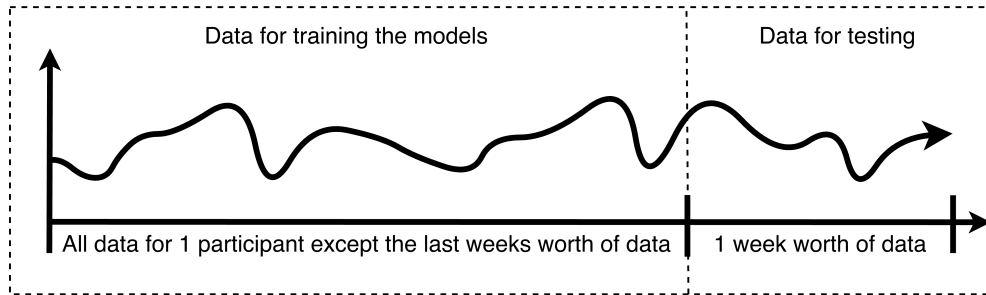


Figure 3.10: The splitting of a participant's data to create training and test sets

3.6.3 Model Selection and Training Approaches

A long short term memory model (LSTM) network architecture based on van Doorn et al.³⁵ was used (two LSTM layers followed by a dense layer). All the training data was scaled to be within the range of zero to one. Hyperparameters were optimized once prior to the repeated model development using grid search and data from all participants except the last week. The gridsearch included the number of neurons in the LSTM layers, activation function, dropout rate, batch size, learning rate, learning rate decay, and early stopping patience. The selected hyperparameter configuration was subsequently fixed and applied to all LSTM models. The complete hyperparameter tuning procedure and implementation are available in the accompanying GitHub repository.⁸³

Four training approaches were used when the models for each participant were developed as described below and illustrated in Figure 3.11. Three LSTM models and the last observation carried forward (LOCF) as a naive baseline. 1. **LOCF**: In the LOCF model, a given CGM measurement (x_4) was used to predict blood glucose values 60 minutes in advance (x_8) in the test set by assuming no change in the last 60 minutes. 2. **Base-Individual**: In the Base-Individual model, each participant's complete CGM data (except for the last week, which was used for the testing data) was used for training. 3. **Base-Generalized**: Here, for each participant, the 11 training corresponding datasets described above were used to generate 11 separate models, which were all subsequently tested on the given participant's last week of CGM data. 4. **Transfer Learned**: This model mirrored the Base-Generalized Model's setup. However, the 11 trained models for each participant were subsequently tailored to the given participant's complete CGM data (except for the last week, which was used for the testing data). The transfer learning technique used is described as transfer 2 in Figure 1.2.

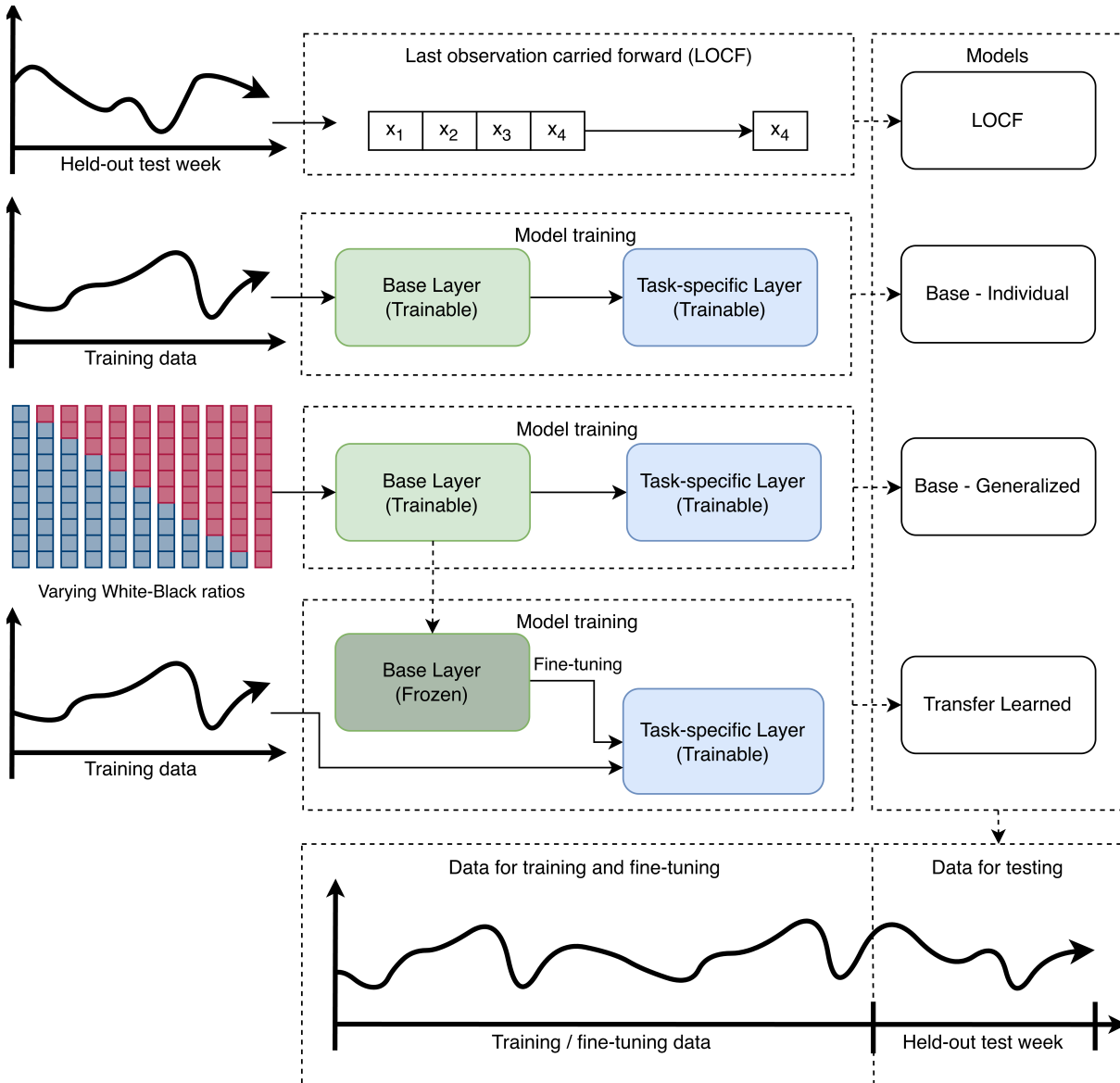


Figure 3.11: The four training approaches. The Base-Individual, Base-Generalized, and Transfer Learned models were based on an LSTM architecture. The base layer (green) consisted of two LSTM layers, while the task-specific layer (blue) consisted of a single dense layer.

3.6.4 Model Evaluation

To evaluate model performance, each participant's last week's data (672 measurements) was reserved for internal testing (Figure 3.10). Model accuracy was assessed by calculating the root mean squared error (RMSE) for each participant across all four modeling approaches. For the LOCF and Base-Individual models, this yielded one RMSE value for each of the 101 White participants and 104 Black participants. For the Base-Generalized and Transfer Learned models, RMSEs were calculated for each participant across 11 data subsets, each representing a unique proportion of White and Black participants. Performance was summarized as the mean RMSE with corresponding 95% confidence intervals (CIs) for each racial group. To evaluate the predictive performance across different training subset compositions for the Base-Generalized and Transfer Learned models, an LMEM was fitted because each participant contributed multiple RMSE values from training subsets with varying proportions of White and Black participants. Participant-specific random intercepts were included to account for the repeated-measures structure. Fixed effects included the proportion of White participants in the training data, participant race, age, sex, training data sample size, and their interaction terms. It was further assumed that the varying proportions of Black and White participants in the training data subsets would affect the performance and an interaction term was therefore added. The interaction term allowed for different slopes in the LMEM which allowed us to test the hypothesis. It was hypothesized that when plotting the proportion of White participants in the training data against the RMSE, the slope would be negative for White participants and positive for Black participants (Figure 3.12). This would indicate that as the proportion of White participants in the training data increased, the RMSE decreased for White participants and increased for Black participants, indicating racial bias in the model. The Base-Individual model would perform the worst since it was trained on the least amount of data and the Transfer Learned model would remove the slopes that were expected to be seen in the Base-Generalized models.

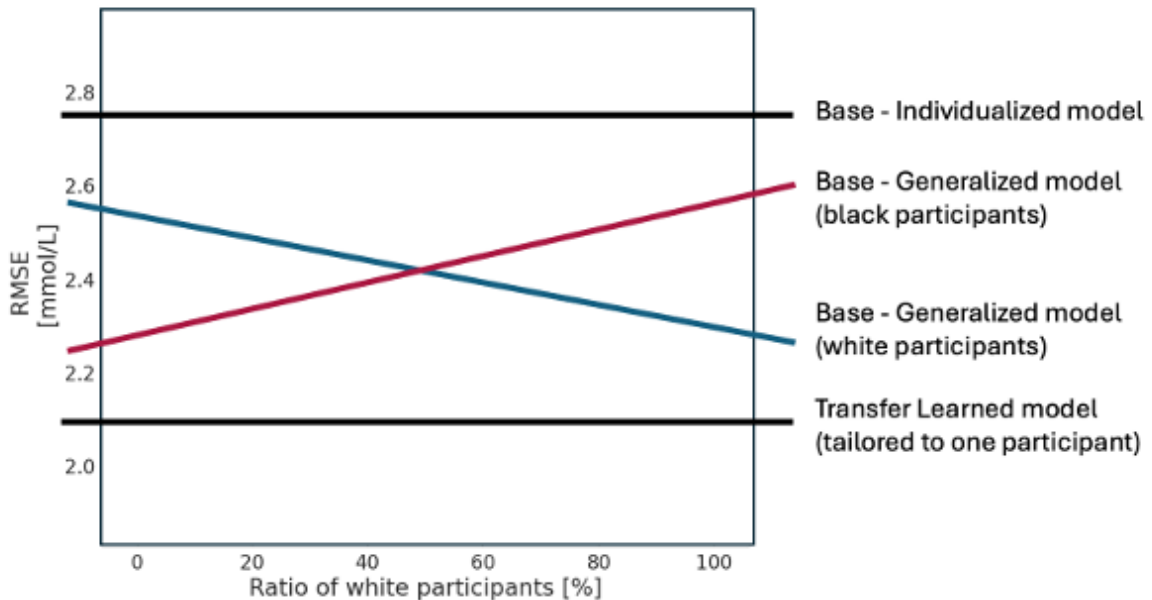


Figure 3.12: The hypothesis for the results of the linear mixed effect models.

Surveillance error grids (SEG) were used to get a general idea of the clinical risk of the prediction models.⁸⁴ The error grids are originally developed for the regulatory bodies to assess clinical accuracy of blood glucose monitoring in post-surveillance before the wide spread use of CGM devices.⁸⁵ and the results should therefore be interpreted with caution when using CGM data. The SEG is a 600x600 grid assigned an individual risk score to each datapoint with 15 colored zones, each indicating severity of a monitoring error that is acted upon in regards to risk of

hypoglycemia or hyperglycemia (Figure 3.13).⁸⁵ To facilitate interpretation, the 15 surveillance error grid zones were collapsed into the five broader risk zones (“No”, “Slight”, “Moderate”, “High”, and “Extreme”), as previously suggested in the literature.⁸⁴ Predictions outside of 0–33 mmol/L were changed to 0 or 33 mmol/L to keep within the upper and lower bounds of the surveillance error grid.

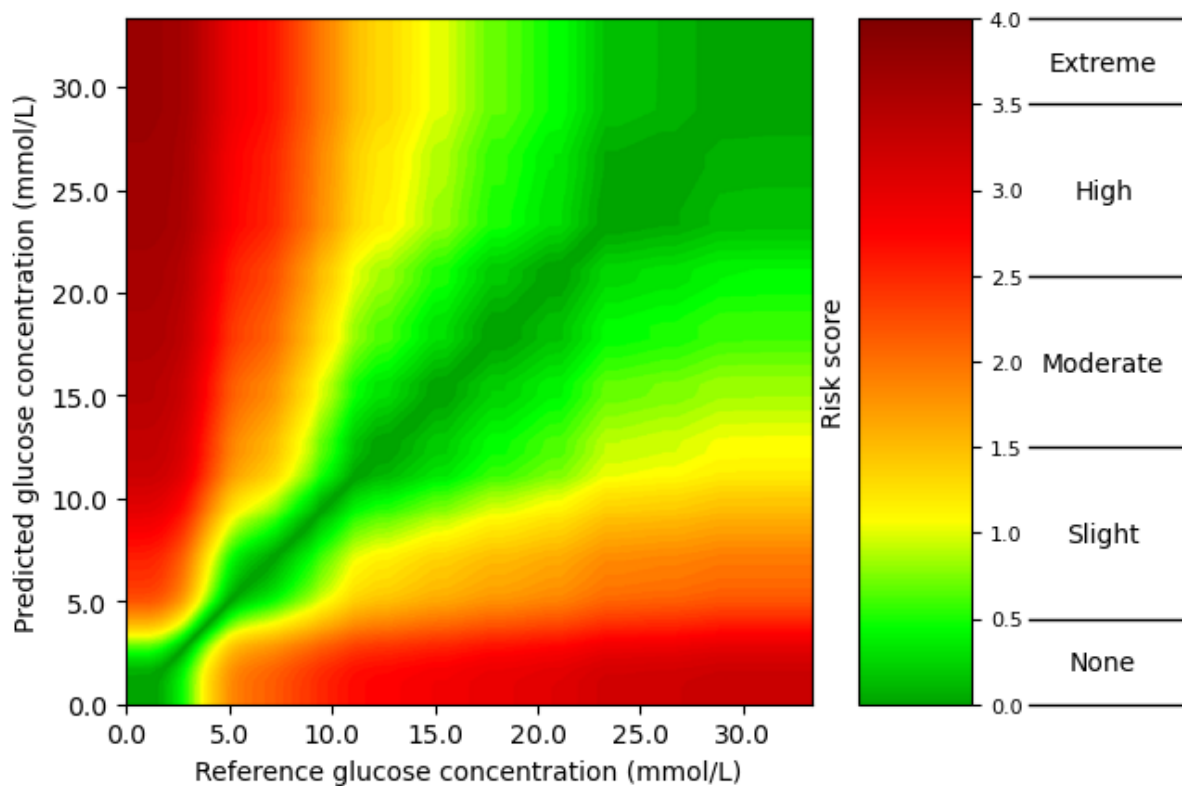


Figure 3.13: Surveillance error grid showing the clinical risk associated with prediction errors. The green region indicates no clinical risk, whereas yellow, orange, and red regions represent increasing levels of clinical risk.

4.1 Scoping Review Results

4.1.1 Study Selection

The study selection process is summarized in Figure 4.1. The Medline and Embase search resulted in 4884 records screened on title and abstract, and 81 reports were assessed for eligibility based on full-text review by two independent reviewers. The citation search on the included 51 reports resulted in 2 additional reports, which finally resulted in 53 included reports in the review. All included and excluded reports are reported in Appendix B.

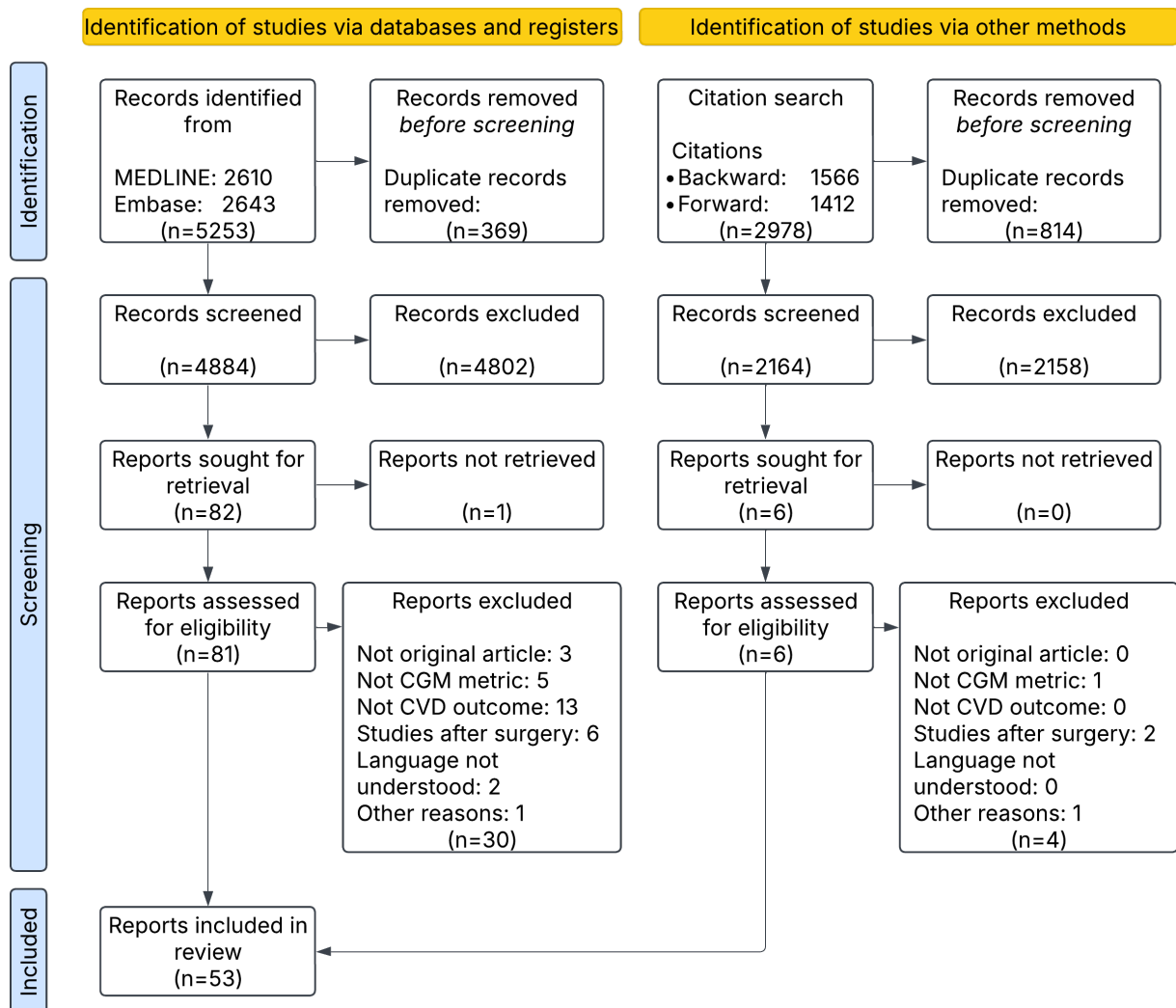


Figure 4.1: Flow diagram of study selection

4.1.2 Study Populations

The study population most common was people with T2D (34 of studies), then people with T1D (20 of 53) and lastly prediabetes (4 of 53). Most studies included adult participants, whereas seven focused on children younger than 18 years. Two studies included only male participants, whereas the remaining studies included both male and female participants.

The geographic distribution was found based on the hospitals and centers where data was collected or by the first author's affiliation when the other option was not available. The distribution was uneven as shown in Figure 4.2. Most studies were conducted in Asia (27 of 53) or Europe (22 of 53). Three studies included data from North America, one was conducted in Australia, and only one included data from an African country, although its study population was predominantly White.^{Borg2011?} No data was included from South America (Appendix C).

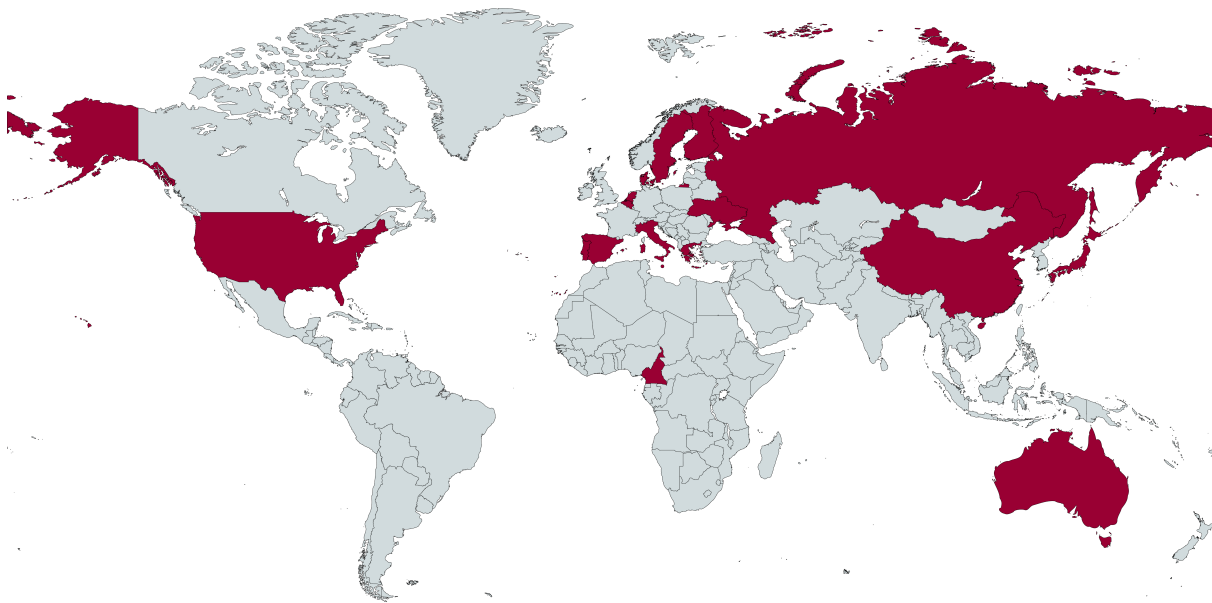


Figure 4.2: Geographical location of the studies. Countries colored in red indicate the geographical locations where the studies were conducted. Map was created using MapChart <https://www.mapchart.net/>.

4.1.3 Study Designs and Demographics

Only one study was a randomized controlled trial,^{33?} and the remaining studies were observational. Most studies used a cross-sectional design, with the cardiovascular outcome assessed at the same time as the CGM sensor was given. Only six studies performed longitudinal analyses. Study sample sizes varied considerably, ranging from 17^{34?} to 6,225^{18?} participants, with a median of 152. CGM data were obtained either from participants' own devices (11 studies) or from CGM devices provided as part of the study (42 studies). Statistical analyses most commonly included linear regression (22 studies) and logistic regression (22 studies), followed by Spearman or Pearson correlation analyses (11 studies) and Cox proportional hazards regression (3 studies). Several studies also performed additional exploratory analyses, such as correlation analyses, to identify relevant covariates for inclusion in regression models.

All included studies primarily investigated associations between CGM-derived measures and cardiovascular outcomes. Only three studies reported predictive performance measures as part of secondary analyses,^{Koroleva?} and none applied machine learning methods. Furthermore, none of the included studies provided openly available data or source code. Instead, availability was either not addressed or limited to access upon reasonable request from the corresponding author.

4.1.4 Continuous Glucose Monitoring

- Add a row “Number of CVD outcomes” and information to the text as well - only if time permits

Medtronic CGM devices were the most frequently used (28 of 53 studies), followed by Abbott (9 of 53) and Dexcom (5 of 53). Other manufacturers included Menarini, Meiqi Company, and SIBIONICS. Sampling intervals ranged from 3 to 15 minutes but were reported in only a minority of studies (21 of 53). TIR (22 studies) and MAGE (23 studies) were the most frequently investigated CGM-derived metrics (Table 4.1) When stratified by T1D and T2D then TIR was the most frequently investigated metric in T1D populations (10 of 23 studies), whereas MAGE was less commonly investigated (5 of 22 studies). In T2D populations, TIR and MAGE were investigated in 11 of 23 and 14 of 22 studies, respectively. Studies conducted in T2D populations also tended to include larger sample sizes than those in T1D populations (median [IQR]: TIR, 510 [405–600] vs. 214 [119–547]; MAGE, 251 [89–411] vs. 57 [30–215]).

Table 4.1: Summary of the frequency of the seven most used CGM-derived metrics

CGM-derived Metrics:	TIR	MAGE	MBG	SD	CV	TBR	TAR
Number of Studies, n	23	22	21	19	19	16	15
T1D, n	10	5	9	7	8	9	8
T2D, n	11	14	8	10	8	7	7
Mixed*	2	3	4	2	3	0	0
Total study size for the CGM-derived metrics**	445 [157-577]	156 [63-344]	105 [71-427]	267 [77-480]	342 [98-577]	123 [77-312]	152 [77-425]
T1D**	214 [119-547]	57 [30-215]	105 [31-347]	267 [54-515]	152 [91-515]	124 [102-161]	138 [98-188]
T2D**	510 [405-600]	251 [89-411]	405 [77-499]	371 [95-484]	499 [358-632]	121 [75-499]	405 [75-499]

* Mixed includes both T1D, T2D and prediabetes

** Total study size for each CGM-derived metrics includes the study populations in all the studies using the CGM-derived metric in the column, reported as median [IQR]

Studies may contribute to more than one CGM-derived metric because multiple metrics could be evaluated within the same study.

Abbreviations: T1D: type 1 diabetes, T2D: type 2 diabetes, MAGE: mean amplitude of glycemic excursions, TIR: time in range, MBG: mean blood glucose, SD: standard deviation, CV: coefficient of variation, TBR: time below range, TAR: time above range.

4.1.5 Prediction studies

Only three studies evaluated the predictive performance of CGM-derived metrics for cardiovascular outcomes, all using logistic regression models with MAGE as the predictor. Two studies reported AUROC values of 0.61 and 0.62, respectively,^{43?} and both found MAGE to outperform HbA1c (AUROC: 0.55 and 0.58, respectively). The third study used receiver operating characteristic analysis to determine the optimal threshold for dichotomizing MAGE during variable selection but did not report the AUROC.^{44?} Instead, a threshold of MAGE 3.4 mmol/L was identified as a risk factor for stenosis and/or occlusion, with a sensitivity of 0.60 and a specificity of 0.61.

4.1.6 Association studies

- ☐ I am thinking of changing table Table 4.2 to be more aligned with the overview tables (?@sec-sec-suppl-subclin-results) An overview of the different outcomes found in the literature both clinical and subclinical are shown in Table 4.2.

Table 4.2: Summary of established clinical and subclinical outcomes and corresponding abbreviations identified in the literature.

Clinical	Subclinical
CVD = Cardiovascular disease	ABI = Ankle-brachial index
- Macrovascular complications	BP = Blood pressure
- Diabetic macroangiopathy	Carotid artery distensibility = Carotid artery distensibility or carotid distensibility coefficient
CAD = Coronary artery disease	CCA = Common carotid artery
- CHD = Coronary heart disease	CIMT = Carotid artery intima-media thickness
- MI = Myocardial infarction	DBP = Diastolic blood pressure
- Heart attack	FMD = Flow-mediated dilation of the brachial artery
- Heart failure	GSM = Gray-scale median of the carotid arteries
- (Unstable) angina pectoris	HRV = Heart rate variability
Cardiovascular events	IMT = Intima-media thickness
- Ischemic heart disease (IHD)	PWV = Pulse wave velocity
LEAD = Lower extremity artery disease	- ba-PWV = Brachial-ankle pulse wave velocity
MACE = Major adverse cardiovascular events	- cr-PWV = Carotid-radial pulse wave velocity
PAD = Peripheral artery disease	- cf-PWV = Carotid-femoral pulse wave velocity
Stroke = Stroke, Cerebrovascular disease, or Cerebrovascular accident	- Aortic PWV = Aortic pulse wave velocity
	SBP = Systolic blood pressure

4.1.7 Clinical Association Studies

- ☐ Does anyone have a suggestion on how to present this data in a table that is smaller than this Table 4.4?
- ☐ Based on AAI comments, then I am removing the columns “Diabetes duration” and “HbA1c, mmol/mol” and will try my best to let studies only take up one row.

Most studies focused exclusively on subclinical cardiovascular outcomes (37 of 53 studies, 70%), whereas 13 studies (25%) investigated only clinical cardiovascular outcomes and three studies (6%) included both clinical and subclinical outcomes. Because the subsequent chapters of this dissertation focus on clinical cardiovascular outcomes, only these findings are discussed in detail below (Table 4.4 and Table D.1). A summary of studies investigating subclinical cardiovascular outcomes is provided in Appendix D.

Table 4.3: Study design and demographics of the included clinical cardiovascular outcome studies

Author, year	Study Design	Population	Size	Age (Years)	Diabetes duration (Years)	HbA1c, mmol/mol (%)	CGM duration
Chen, 2020 [†]	Longitudinal (prospective) and cross-sectional (retrospective) FU: in-hospital or within 3 months after discharge from hospital	T2D with CAD (Only male) BG control, n = 90 BG fluctuation, n = 120	210	BG controls: 55.53 ± 7.30 BG fluctuation: 56.41 ± 7.67	BG controls: 6.59 ± 2.30 BG fluctuation group: 6.92 ± 2.25	nr	2 days
He, 2023	Longitudinal (prospective) FU: 1 year	T2D with kidney disease on hemodialysis High TIR, n = 12 Low TIR, n = 15	27	High TIR: 66 [63, 73] Low TIR: 70 [64, 75]	High TIR: 2 [1.7, 10] Low TIR: 5 [0.75, 20]	High TIR: 43 [37, 51] mmol/mol (6.1 [5.5, 6.8]%) Low TIR: 66 [46, 70] mmol/mol (8.2 [6.4, 8.6]%)	14 days
Lu, 2021	Longitudinal (prospective) FU: until death occurred or 3–13 years Median: 6.9 years	T2D, hospitalized	6225	mean age = 61.7 years	mean: 9.7 years	74.0 ± 24.0 mmol/mol (8.9 ± 2.2%)	72 hours
Wei, 2019	Longitudinal (prospective) Median FU [IQR]: 31 months (22, 56)	T2D Divided into three groups: 1) No hypo, n = 1173 2) Mild hypo (lv1), n = 323 3) Severe hypo (lv3), n = 24	1520	No hypoglycemia: 58.59 ± 11.26 Hypoglycemia: 62.27 ± 11.58	No hypoglycemia: 6.46 ± 6.00 Hypoglycemia: 7.78 ± 7.37	No hypoglycemia: (8.19 ± 2.10%) Hypoglycemia: (7.73 ± 1.96%)	3 days

Table 4.3: Study design and demographics of the included clinical cardiovascular outcome studies

Author, year	Study Design	Population	Size	Age (Years)	Diabetes duration (Years)	HbA1c, mmol/mol (%)	CGM duration
Bezerra, 2023	Cross-sectional	T1D	161	37.4 ± 13.4	17.7 ± 10.6	$(7.5 \pm 1.1\%)$	14 days
De Meulemeester, 2024 [†]	Cross-sectional	T1D	808	44.8 ± 15.2	23.1 ± 13.6	63 ± 13 mmol/mol ($7.9 \pm 1.2\%$)	2 weeks
Deng, 2023	Cross-sectional	T2D	860	HP 1 carriers: 53.5 ± 13.3 HP 2-2: 51.7 ± 14.6	HP 1 carriers: 8.6 ± 6.4 HP 2-2: 8.6 ± 6.7	HP1 carriers: 73.0 ± 24.0 mmol/mol ($8.8 \pm 2.2\%$) Hp2-2: 72.0 ± 23.0 mmol/mol (8.7 ± 2.1)	3 days
El Malahi, 2022	Cross-sectional	T1D starting on sensor-augmented pump therapy	515	Mean $\pm$ SD [years]: 42.2 ± 12.5	22.3 ± 11.6	60 ± 9.8 mmol/mol ($7.6 \pm 0.9\%$)	2 weeks
Guo, 2021	Cross-sectional	T1D or T2D with atrial fibrillation With Stroke, n = 48 Without Stroke, n = 462	510	Stroke: 70.3 ± 12.1 No stroke: 68.1 ± 9.4	nr	No Stroke: 7.4 ± 2.1 Stroke: 8.2 ± 1.7	72 hour
Li, 2020	Cross-sectional	T2D with LEAD, n = 179 T2D without LEAD, n = 157	336	Without LEAD: 55.94 ± 12.45 With LEAD: 65.56 ± 11.99	Without LEAD: 6.92 ± 3.54 With LEAD: 10.32 ± 4.14	Without LEAD: $7.85 \pm 1.41\%$ With LEAD: $8.97 \pm 1.63\%$	72 hours

Table 4.3: Study design and demographics of the included clinical cardiovascular outcome studies

Author, year	Study Design	Population	Size	Age (Years)	Diabetes duration (Years)	HbA1c, mmol/mol (%)	CGM duration
Magri, 2018 [†]	Cross-sectional	T2D	121	64 [57, 68]	3 [2, 5]	median: 45 mmol/mol (6.8%)	72 hours
Mi, 2012	Cross-sectional	T2D with chest pain Without CAD, n = 202 With CAD, n = 84	286	Without CAD: 62.8 ± 8.7 With CAD: 66.6 ± 9.2	nr	Non-CAD: 7.51 ± 0.80% CAD: 7.75 ± 0.92%	72 hours (Only intermediate 48 hours were used)
Sheng, 2023	Cross-sectional	T2D, hospitalized	545	mean age of (61.22 ± 11.21) years	nr	8.51 ± 1.85%	7-14 days
Su, 2011	Cross-sectional	T2D with chest pain Without CAD, n = 92 With CAD, n = 252	344	Without CAD: 61 ± 9 With CAD: 65 ± 9	Without CAD: 4.8 ± 5.7 With CAD: 6.5 ± 6.4	non-CAD: 7.5 ± 1.4% CAD: 7.6 ± 1.5%	72 hours (only 48 hours used)
Watanabe, 2017	Cross-sectional	Prediabetes, hospitalized	28	64.3 ± 12.8	nr	5.41 ± 0.35%	72 hours (Only used the middle 48 hours)

Zhang, 2013	Cross-sectional	T2D with cardiovascular complications Group A: healthy individuals Group B: T2D without cardiovascular complications Group C: T2D with cardiovascular complications	92	Group A: 56.3 ± 6.1 Group B: 56.1 ± 6.6 Group C: 61.7 ± 7.2	nr	Group A: 5.3 ± 0.3% Group B: 6.6 ± 1.2% Group C: 7.5 ± 1.4%	72 hours
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† This study appears both in the clinical and subclinical disease outcome tables due to investigating multiple CVD outcomes. Note: Age is reported in years either as an interval, mean ± SD, or median (IQR). Diabetes duration values originally reported in months were converted to years for consistency (months ÷ 12).
Abbreviations: CGM: continuous glucose monitoring, FU: follow-up, T1D: type 1 diabetes, T2D: type 2 diabetes, CAD: coronary artery disease, hypo: hypoglycemia, LEAD: lower extremity arterial disease, IQR: interquartile range, nr: not reported.

- ☐ I need to add a total row of summarized values in Table 4.4
- ☐ I need to add an other column in Table 4.4 for other CGM derived metrics

An overview of the most frequently used CGM-derived metrics stratified based on clinical and subclinical cardiovascular outcomes are shown in Table 4.4. Outcomes related to subclinical cardiovascular outcomes were more frequently studied (59 studies) compared to clinical outcomes (26 studies) and studies were predominantly done among people with T2D. Among clinical cardiovascular outcomes, coronary artery disease was examined most frequently (6 studies), followed by macrovascular complications (5 studies) and the Gensini score (4 studies) and for subclinical cardiovascular outcomes, particularly arterial wall thickness were frequently investigated (19 studies), arterial stiffness (12 studies), cardiac and pulse-related measures (10 studies), and flow resistance (9 studies).

TIR was the most frequently reported CGM-derived metric across most arterial-stiffness outcomes, followed by MAGE, MBG, SD, and CV. MAGE was reported most frequently in studies evaluating coronary artery disease, Gensini score, and Syntax score. By contrast, relatively few CGM-derived metrics were reported for mortality, major adverse cardiovascular events, stroke, peripheral artery disease, and lower-extremity arterial disease.

The outcomes for the 53 studies can be found in Appendix D

Table 4.4: Overview of the diabetes populations and the most frequently reported CGM-derived metrics, stratified by cardiovascular outcome.

Cardiovascular Outcomes	Number of studies	T1D	T2D	Mixed (T1D/T2D)	Prediabetes or impaired glucose	TIR	MAGE	MBG	SD	CV	TBR	TAR
Macrovascular Mortality	2	-	2	-	-	1	-	-	-	-	-	-
Macrovascular Complications	5	3	2	-	-	3	-	-	1	3	2	1
Major adverse cardiovascular events	2	-	2	-	-	-	-	-	-	-	-	-
Coronary Artery Disease	6	1	5	-	-	2	2	-	-	-	-	-
Gensini score	4	-	3	-	1	-	3	-	-	-	-	-
Syntax score	2	-	1	-	1	-	2	-	-	-	-	-
Stroke	3	1	1	1	-	2	-	-	-	-	-	-
Peripheral Artery Disease	1	1	-	-	-	1	-	-	-	-	-	-
Lower extremity arterial disease	1	-	1	-	-	1	-	-	1	1	-	-
Arterial stiffness	12	6	5	1	-	8	6	7	6	5	4	4
Flow resistance	9	4	4	1	-	3	5	6	3	4	3	3
Arterial wall thickness	19	6	11	1	1	8	9	10	9	10	5	4
Arterial wall composition	6	2	4	-	-	3	2	5	2	3	5	5
Cardiac and pulse related measures	10	2	6	2	-	4	1	5	7	3	2	2
Arterial lumen	3	2	1	-	-	2	1	1	-	1	1	1

Cardiovascular outcomes were grouped into clinically meaningful categories. A detailed description of the outcome categories is provided in supplementary. Values represent the number of included studies evaluating each CGM-derived metric for the specified cardiovascular outcome. A study could contribute to more than one cell if multiple CGM-derived metrics were investigated.

Abbreviations: *TIR*, time in range; *MAGE*, mean amplitude of glycemic excursions; *MBG*, mean blood glucose; *SD*, standard deviation; *CV*, coefficient of variation; *TBR*, time below range; *TAR*, time above range

Table 4.5: Overview of adjustments used in the association studies

Adjustment Domain	Clinical	Subclinical	Total
Demographic	14	22	36
Anthropometric	10	19	29
Lifestyle	8	12	20
Medical History	14	18	32
Physiological	7	17	24
Biochemical	11	21	32
Medication	7	10	17
CGM/GV	5	9	14
CVD marker	1	5	6
Not applicable	3	9	12
Not reported	1	5	6

A study could contribute to more than one adjustment domain.

4.2 Baseline Characteristics Tables of the Datasets

The CANCAN dataset presented below was used in Study II (Table 4.6). The total number of included participants were 159 and 67 (42%) had elevated NT-proBNP values. Participants with elevated NT-proBNP tended to be older, have a longer duration of diabetes, and more frequently have cardiovascular disease. They also tended to have lower BMI and lower LDL and total cholesterol concentrations, while exhibiting modestly better overall glycaemic control but slightly higher glycaemic variability.

Table 4.6: Baseline table for CANCAN data.

Variable	Total	NT-proBNP \$ \$125 pg/mL	NT-proBNP >125 pg/mL
N	159	92	67
Age (years)	65 [56, 71]	60 [53, 68]	68 [64, 76]
Sex			
– Male	99 (62%)	55 (60%)	44 (66%)
– Female	60 (38%)	37 (40%)	23 (34%)
Smoking status			
– Never or former	135 (85%)	77 (84%)	58 (87%)
– Current	24 (15%)	15 (16%)	9 (13%)
Diabetes duration (years)	16 [11, 24]	14 [9, 22]	19 [14, 25]
SCORE2-Diabetes (%)	14.8 [9.1, 20.9]	11.9 [7.8, 17.2]	19.0 [13.4, 25.4]
HbA1c (mmol/mol)	64 [55, 80]	66 [56, 86]	60 [54, 72]
Systolic blood pressure (mmHg)	132 [123, 143]	132 [123, 143]	131 [122, 142]
HDL (mmol/L)	1.0 [0.9, 1.2]	1.0 [0.9, 1.2]	1.0 [0.9, 1.3]
Total cholesterol (mmol/L)	3.9 [3.2, 4.8]	4.1 [3.5, 4.9]	3.6 [3.0, 4.2]
eGFR (mL/min/1.73 m ²)	57 [39, 74]	68 [52, 82]	48 [27, 63]
CGM-derived metrics			
– CGM days (n/participant)	13 [10, 13]	13 [9, 13]	13 [10, 13]
– Glucose management indicator (%)	7.3 [6.8, 8.4]	7.5 [6.8, 8.6]	7.2 [6.7, 8.0]
– Time in range (%)	59.6 [31.9, 78.9]	58.2 [20.2, 78.6]	62.1 [40.8, 81.0]
– Time below range (%)	0.0 [0.0, 1.5]	0.0 [0.0, 0.8]	0.2 [0.0, 2.0]
– Time above range (%)	36.6 [17.6, 68.1]	41.8 [19.8, 79.5]	33.1 [14.1, 58.2]

Table 4.6: Baseline table for CANCAN data.

Variable	Total	NT-proBNP \$ \$125 pg/mL	NT-proBNP >125 pg/mL
– Coefficient of variation (%)	27.3 [22.5, 31.9]	25.3 [21.2, 30.3]	28.3[24.8, 34.2]
– Mean amplitude of glycemic excursions (mmol/L)	6.3 [5.0, 7.9]	6.0 [4.9, 7.7]	6.7 [5.1, 8.5]

Categorical values are given by number (percentage), and numerical values are given by median [interquartile range].

The T1DX dataset presented below was used in Study II and Study III (Table 4.7). A total of 205 participants were included, of whom 101 (49%) were White. Compared with Black participants, White participants tended to have lower HbA1c levels (63 mmol/mol vs. 72 mmol/mol) and a higher proportion of children (49% vs. 32%). Females constituted the majority of both racial groups (46% and 40%).

Table 4.7: Baseline table for T1DX data.

Characteristics	Total	Race: White	Race: Black
N	205	101	104
Race (White)	101 (49%)	101 (100%)	0 (0%)
Sex (male)	88 (43%)	46 (46%)	42 (40%)
Age (years)	28 [15, 47]	22 [13, 46]	32 [16, 47]
Age group (child, < 18 yrs)	82 (40%)	49 (49%)	33 (32%)
Age at T1D diagnosis (years)	11 [7, 22]	9 [6, 16]	14 [8, 25]
BMI (kg/m ²)	25.1 [21.7, 29.2]	24.0 [20.7, 28.6]	26.7 [22.3, 30.0]
HbA1c (mmol/mol)	67 [57, 78]	63 [55, 73]	72 [60, 85]
CGM-derived metrics			
– CGM days (n/participant)	58 [46, 69]	63 [47, 71]	56 [45, 66]
– Glucose management indicator (%)	7.5 [7.0, 8.2]	7.4 [6.9, 7.9]	7.7 [7.0, 8.5]
– Time in range (%)	47 [35, 57]	49 [39, 60]	42 [31, 54]
– Time above range (%)	45 [30, 56]	42 [28, 53]	48 [32, 60]
– Time below range (%)	8 [4, 14]	6 [4, 10]	11 [5, 15]
– Coefficient of variation (%)	43.3 [39.6, 48.6]	42.2 [38.6, 46.5]	46.7 [40.4, 51.9]
– Mean amplitude of glycemic excursions (mmol/L)	9.9 [8.2, 12.1]	9.6 [8.1, 11.2]	10.6 [8.5, 13.0]

Categorical values are given by number (percentage), and numerical values are given by median [interquartile range].

4.3 Foundation Model Embedding Analysis

4.3.1 Cosine Distances

- ☐ Add a very brief text to the figures
- ☐ fig-cosim_MAGE needs to be changed, the change in resolution messed up the figures + colors needs to be checked to make sure they are in the right order
- ☐ AAI suggests to trim figure text, needs to allocate time to think about that

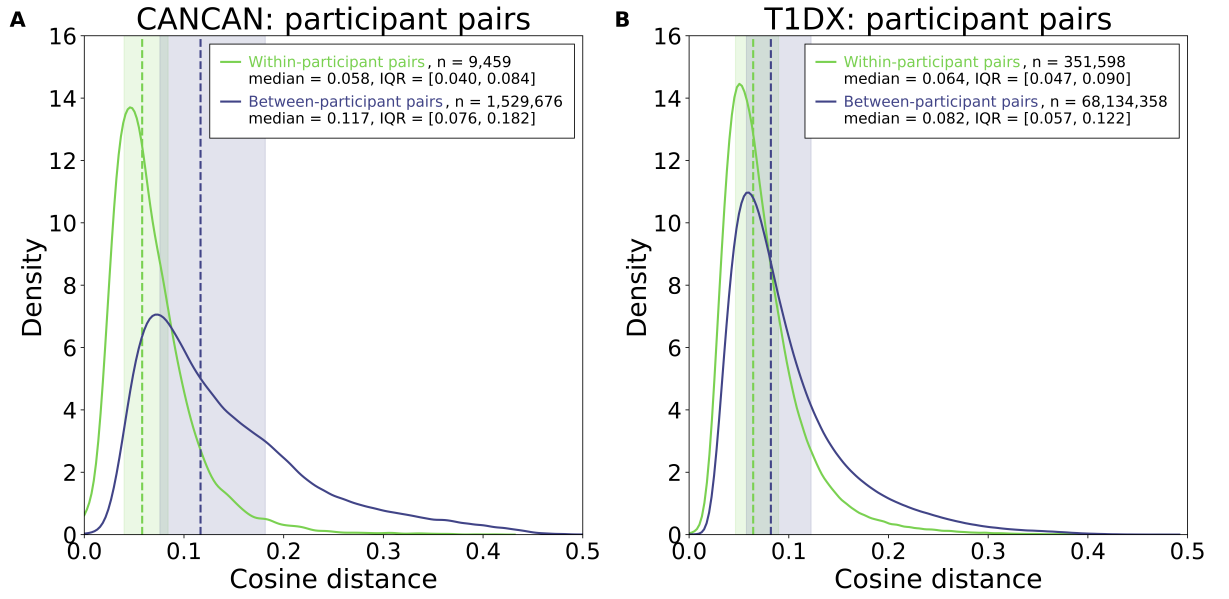


Figure 4.3: Kernel density plots of cosine distances for within- and between-participants in the CANCAN dataset (A) and the T1DX dataset (B). With the median shown as a vertical dashed line and the interquartile range shown as a shaded area around the median.

4.3.2 Embeddings and Glycemic Control

□ Need to add some text in this section

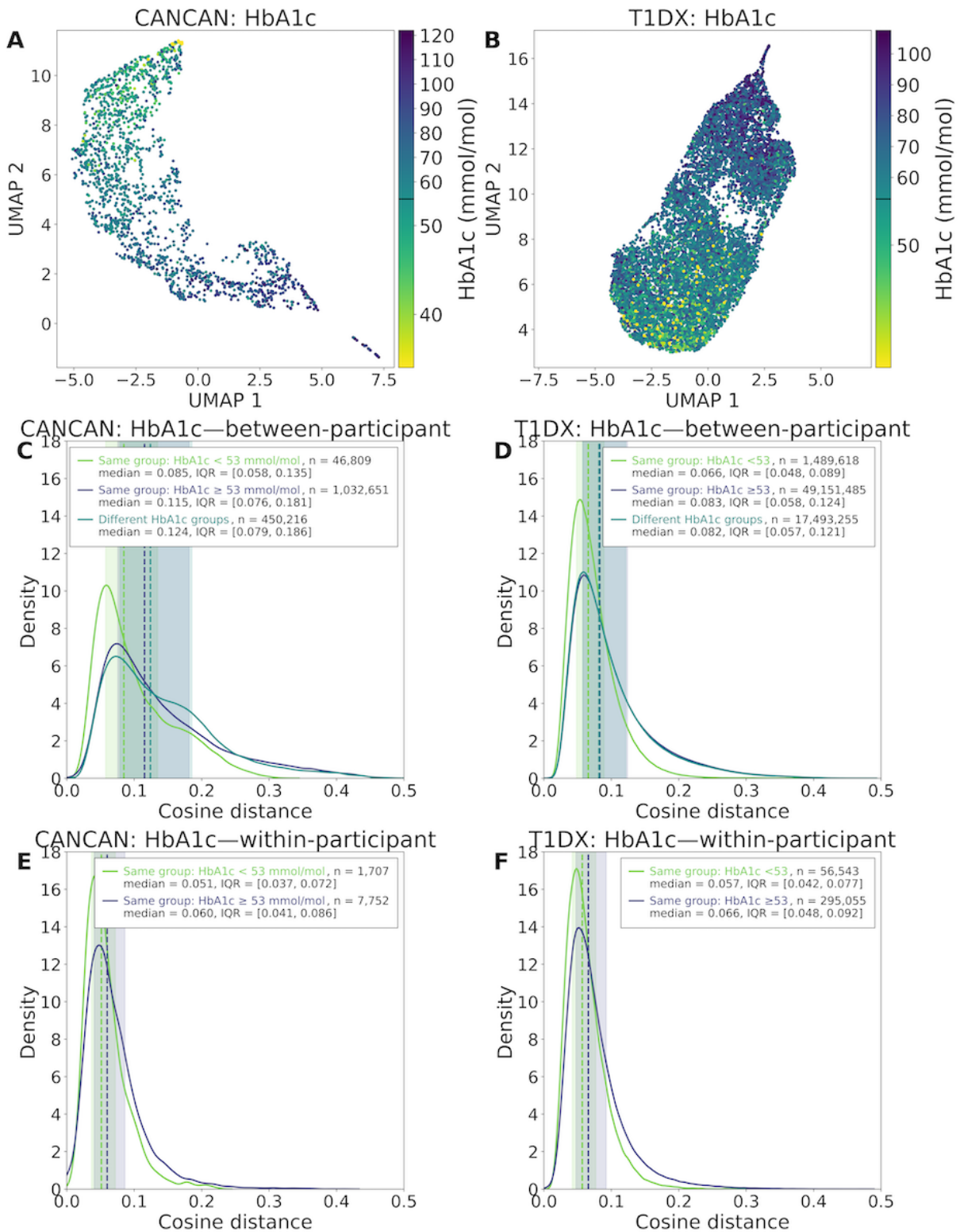


Figure 4.4: UMAPs and kernel density plots over cosine distance pairs for within- and between-participants grouped based on glycemic control metrics for the CANCAN dataset. The UMAPs were colored based on clinical characteristics, with each point in the UMAP representing a single day, colors are assigned to each day at the participant level; thus, the colors reflect overall participant-level rather than day-specific values. The vertical dashed line in the density plots are the medians with the shaded area around the median being interquartile ranges.

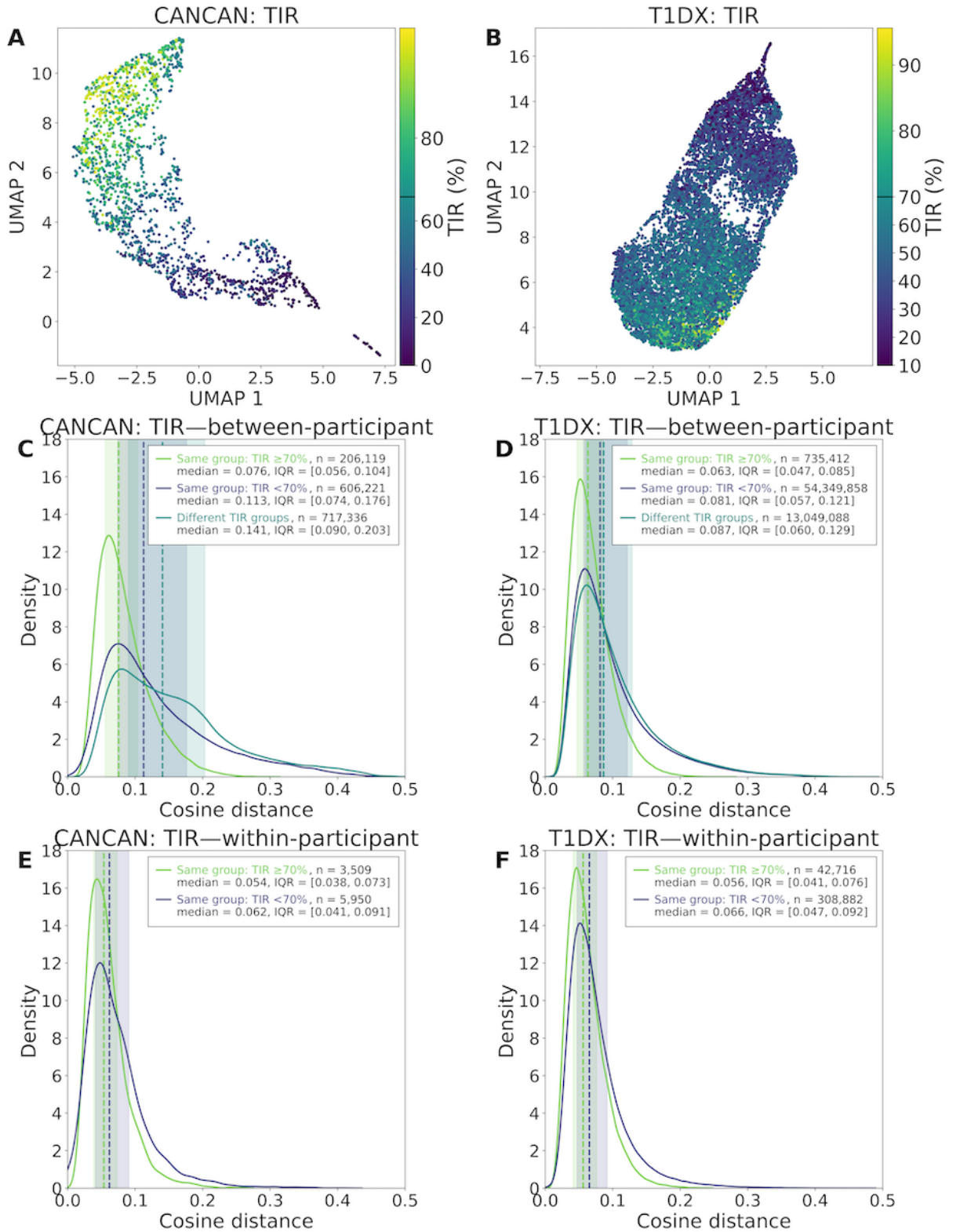


Figure 4.5: UMAPs and kernel density plots over cosine distance pairs for within- and between-participants grouped based on glycemic control metrics for the T1DX dataset. The UMAPs were colored based on clinical characteristics, with each point in the UMAP representing a single day, colors are assigned to each day at the participant level; thus, the colors reflect overall participant-level rather than day-specific values. The vertical dashed line in the density plots are the medians with the shaded area around the median being interquartile ranges..

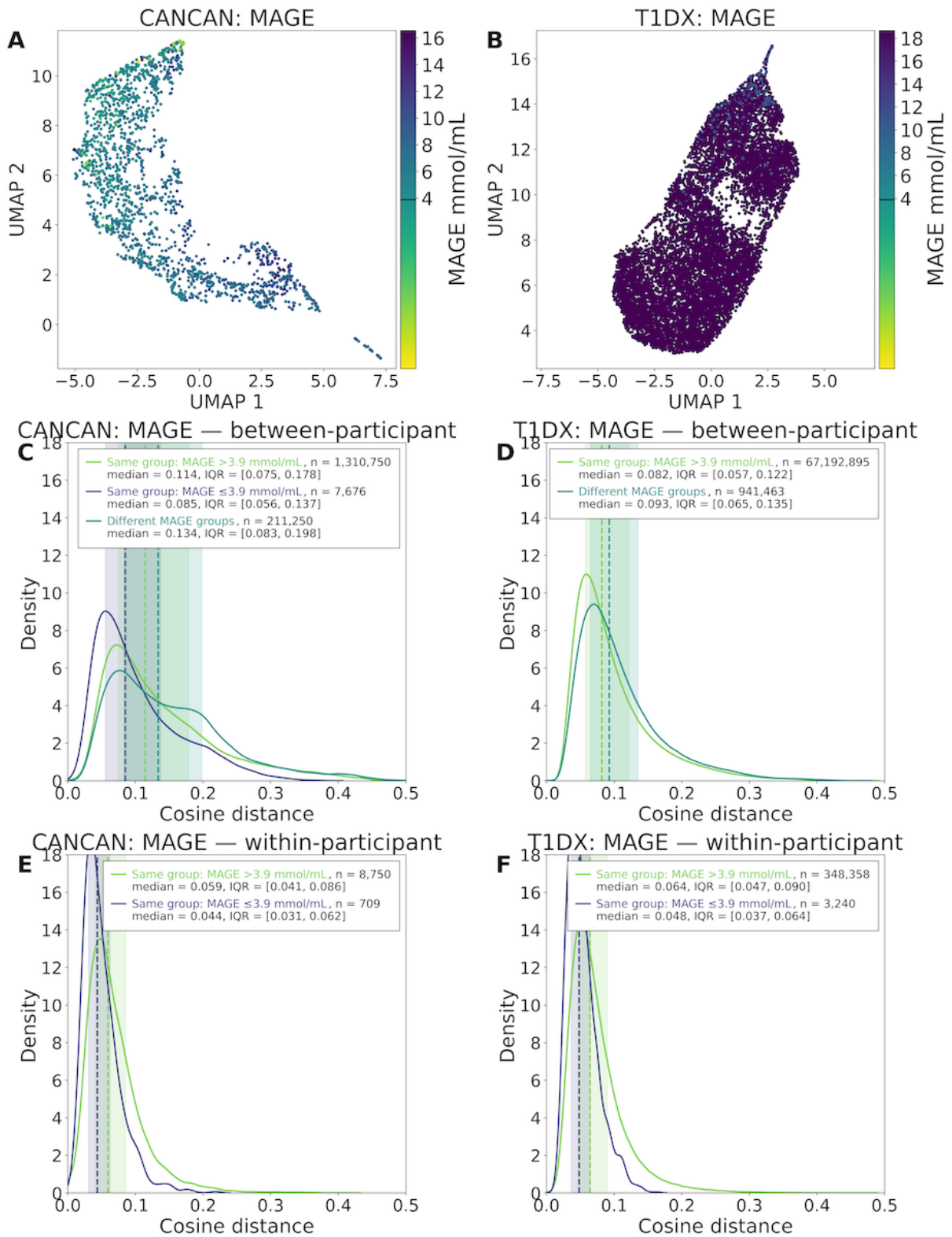


Figure 4.6: UMAPs and kernel density plots over cosine distance pairs for within- and between-participants grouped based on MAGE for the two datasets. The UMAPs were colored based on clinical characteristics, with each point in the UMAP representing a single day, colors are assigned to each day at the participant level; thus, the colors reflect overall participant-level rather than day-specific values. The vertical dashed line in the density plots are the medians with the shaded area around the median being interquartile ranges..

4.3.3 Embeddings and NT-proBNP in CANCAN and race in T1DX

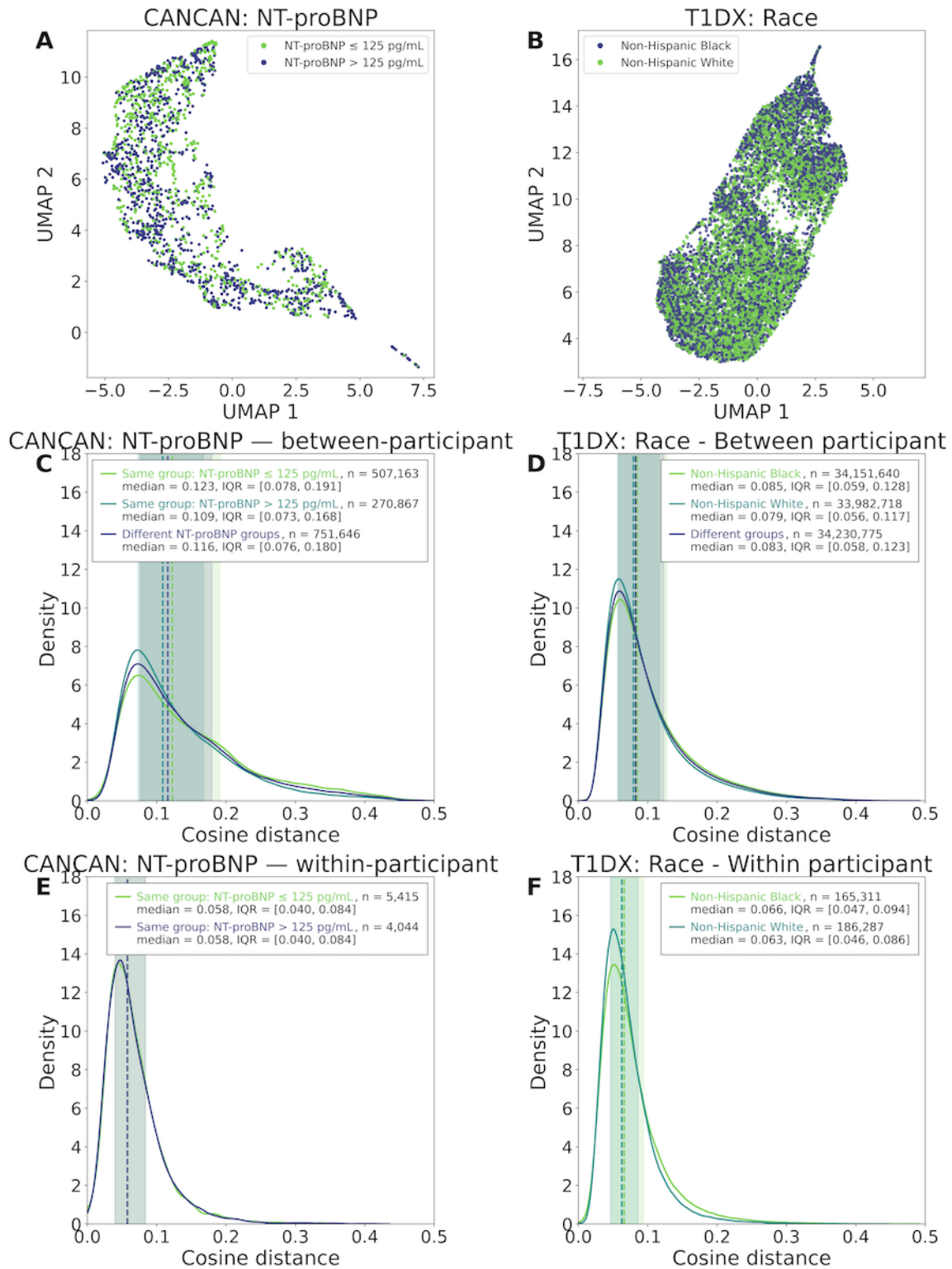


Figure 4.7: UMAPs and kernel density plots over cosine distance pairs for within- and between-participants grouped based on CVD and NT-proBNP for the CANCAN dataset. The UMAPs were colored based on clinical characteristics, with each point in the UMAP representing a single day, colors are assigned to each day at the participant level; thus, the colors reflect overall participant-level rather than day-specific values. The vertical dashed line in the density plots are the medians with the shaded area around the median being interquartile ranges..

4.4 Prediction of NT-proBNP Results

Results from the prediction analysis are shown in Table 4.8. The highest optimism-corrected AUROC was achieved by Model A2 which included both all nine clinical variables from the SCORE2-Diabetes input and the CGM-derived metrics TIR and MAGE (AUROC: 0.81 [0.77–0.88]). Optimism-corrected AUROCs for the models with fewer predictors had modest performance (AUROC: 0.72 to 0.77), however the optimism-correction was smaller and the prediction stability was more stable the fewer predictors were included in the models (Models: B1, B2, and B3) (@#sec-suppl-pred-stability). The models including only CGM-data either as derived metrics (Model C1) or as UMAP dimensions (Model D1a, D1b, D1c) performed the worst (AUROC: 0.53 to 0.69). Age seemed a better predictor for elevated NT-proBNP (AUROC: 0.73 [0.65 - 0.80])

Table 4.8: Area under the receiver operating characteristic curve for the included models.

Models	Corrected [95% CI]	Apparent	Optimism	Number of predictors
Model A0: (Age)	0.73 [0.65 - 0.80]	0.73	0.00	1
Model A1: (SCORE2-Diabetes input)	0.79 [0.76 - 0.87]	0.83	0.04	9
Model A2: (Model A1 + TIR and MAGE)	0.81 [0.77-0.88]	0.86	0.05	11
Model A3a: (Model A1 + UMAP (min))	0.80 [0.75 - 0.87]	0.84	0.05	11
Model A3b: (Model A1 + UMAP (max))	0.79 [0.75 - 0.87]	0.84	0.05	11
Model A3c: (Model A1 + UMAP (mean))	0.79 [0.75 - 0.87]	0.84	0.05	11
Model B1: (SCORE2-Diabetes score)	0.74 [0.65 - 0.81]	0.73	0.00	1
Model B2: (Model B1 + TIR and MAGE)	0.77 [0.70 - 0.85]	0.79	0.01	3
Model B3a: (Model B1 + UMAP (min))	0.72 [0.65 - 0.80]	0.74	0.02	3
Model B3b: (Model B1 + UMAP (max))	0.72 [0.65 - 0.81]	0.74	0.02	3
Model B3c: (Model B1 + UMAP (mean))	0.73 [0.65 - 0.81]	0.75	0.02	3
Model C1: (TIR and MAGE)	0.69 [0.61 - 0.77]	0.69	0.01	11
Model D1a: (UMAP, max*)	0.55 [0.48 - 0.64]	0.58	0.03	2
Model D1b: (UMAP, min*)	0.53 [0.43 - 0.63]	0.57	0.04	2
Model D1c: (UMAP, mean*)	0.53 [0.43 - 0.64]	0.56	0.03	2

* Indicates the pooling technique for the embeddings. * Abbreviations: * CI: Confidence Interval, CGM: Continuous Glucose Monitoring, UMAP: Uniform Manifold Approximation and Projection.

4.5 Algorithmic Fairness in Prediction Models

Bias is an important spect to consider when working data drivenly. In study III, we therefore evaluates bias in CGM trained machine learning models for forecasting in the T1DX cohort

4.5.1 Bias in Machine Learning Models

The RMSE of the models are shown in Table 4.9 and Table 4.10. Base-Generalized and Transfer Learned models achieved lower prediction errors (RMSE: 2.00 to 2.06 mmol/L) than the naive last observation carried forward (LOCF)(RMSE: 2.41 and 2.45 mmol/L) and Base-Individual models (RMSE: 2.54 to 2.63 mmol/L) for both White and Black participants (Table 4.9 and Table 4.10).

Table 4.9: Root mean squared errors with 95% CIs for the different models evaluated on White participants.

Ratio	LOCF	Base-Individual	Base-Generalized	Transfer Learned
0			2.04 [1.95, 2.13]	2.02 [1.94, 2.11]
10			2.04 [1.95, 2.13]	2.02 [1.94, 2.11]
20			2.04 [1.95, 2.12]	2.02 [1.94, 2.10]
30			2.03 [1.94, 2.12]	2.02 [1.94, 2.10]
40	2.45 [2.32, 2.57]	2.54 [2.38, 2.70]	2.03 [1.94, 2.12]	2.02 [1.94, 2.10]
50	(independent	(independent	2.03 [1.94, 2.11]	2.02 [1.93, 2.10]
60	of ratio)	of ratio)	2.02 [1.94, 2.11]	2.02 [1.93, 2.10]
70			2.02 [1.93, 2.11]	2.01 [1.93, 2.10]
80			2.02 [1.93, 2.11]	2.01 [1.93, 2.10]
90			2.01 [1.92, 2.10]	2.01 [1.93, 2.10]
100			2.01 [1.92, 2.10]	2.01 [1.92, 2.10]

Values that vary by proportion are estimated using linear mixed-effects models. These models were adjusted to the population's age (40% children) and sex (57% women) distribution.

Abbreviations: LOCF: last observation carried forward.

Table 4.10: Root mean squared errors with 95% CIs for the different models evaluated on Black participants.

Ratio	LOCF	Base-Individual	Base-Generalized	Transfer Learned
0			2.05 [1.96, 2.14]	2.00 [1.91, 2.08]
10			2.05 [1.96, 2.14]	2.00 [1.91, 2.08]
20			2.05 [1.96, 2.14]	2.00 [1.92, 2.08]
30			2.05 [1.97, 2.14]	2.00 [1.92, 2.08]
40	2.41 [2.28, 2.53]	2.63 [2.47, 2.79]	2.05 [1.97, 2.14]	2.00 [1.92, 2.08]
50	(independent	(independent	2.05 [1.97, 2.14]	2.00 [1.92, 2.08]
60	of ratio)	of ratio)	2.06 [1.97, 2.14]	2.00 [1.92, 2.08]
70			2.06 [1.97, 2.14]	2.00 [1.92, 2.08]
80			2.06 [1.97, 2.14]	2.00 [1.92, 2.09]
90			2.06 [1.97, 2.15]	2.00 [1.92, 2.09]
100			2.06 [1.97, 2.15]	2.00 [1.92, 2.09]

<!-- Values that vary by proportion are estimated using linear mixed-effects models. These models were adjusted to the population's age (40% children) and sex (57% women) distribution. *Abbreviations:* LOCF: last observation carried forward. -->

For the Base-Generalized models, prediction performance gradually diverged between White and Black participants as the proportion of White participants in the training data increased (Figure 4.8). For example, when the training data changed from 0% to 100% White participants, the difference in RMSE between the racial groups increased by 0.04 mmol/L [95% CI: 0.001 to

0.008], $p=0.02$, indicating that training data composition affected the two groups differently (Table 4.11). This divergence was not observed after transfer learning (0.002 mmol/L [95% CI: -0.001 to 0.005], $p=0.20$) (Table 4.12). A statistically significant differences were however found in both the Base-Generalized and the Transfer Learned models for age groups (0.482 [0.355 to 0.610], $p < 0.001$, Table 4.11 and 0.411 [0.290 to 0.533], $p < 0.001$, Table 4.12 respectively). The differences in age groups showed a lower error when forecasting blood glucose values for adults compared to children. Larger sample sizes available for transfer learning were associated with lower prediction error ($\beta = -0.111$ mmol/L [-0.161 to -0.061] mmol/L, $p < 0.001$) for every additional 1,000 samples used for fine-tuning (Table 4.12). In contrast, the size of the dataset used to train the Base-Generalized model was not associated with prediction error.

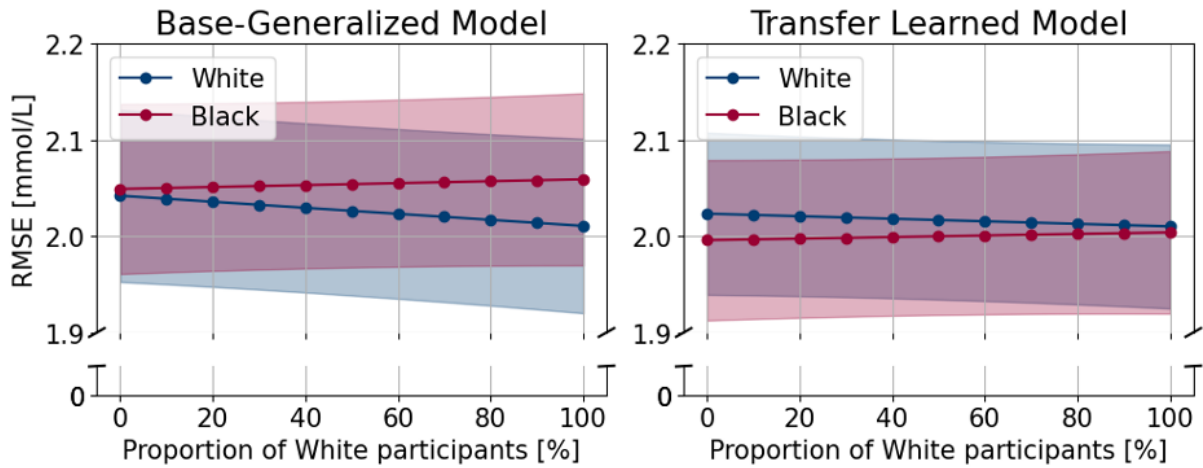


Figure 4.8: Estimated root mean squared errors (RMSE) across 11 different proportions of White and Black participants in the dataset.

Table 4.11: Model coefficients from linear mixed effects models for root mean squared errors of the Base-Generalized models.

Predictor	Coefficient	95% CI	P-value
Intercept	1.893	[1.762, 2.024]	<0.001
Race			
White	Reference		
Black	0.007	[-0.118, 0.132]	0.91
Age			
Adult	Reference		
Child	0.482	[0.355, 0.610]	<0.001
Sex			
Male	Reference		
Female	-0.076	[-0.201, 0.048]	0.23
Training size Base-Generalized*	0.019	[-0.055, 0.093]	0.62
Ratio (per 10%)	-0.003	[-0.007, 0.001]	0.12
Ratio · Race	0.004	[0.001, 0.008]	0.02

* Per 100,000 samples. (Training size was centered at 510,000 as the training data had a mean of 514,293 samples)

Table 4.12: Model coefficients from linear mixed effects models for root mean squared errors of the Transfer Learned models.

Predictor	Coefficient	95% CI	P-value
Intercept	1.890	[1.768, 2.015]	<0.001
Race			
White	Reference		
Black	-0.028	[-0.147, 0.092]	0.65
Age			
Adult	Reference		
Child	0.411	[0.290, 0.533]	<0.001
Sex			
Male	Reference		
Female	-0.057	[-0.173, 0.060]	0.34
Training size Base-Generalized*	0.011	[-0.060, 0.081]	0.77
Training size tl**	-0.111	[-0.161, -0.061]	<0.001
Ratio (per 10 units)	-0.001	[-0.003, 0.001]	0.46
Ratio · Race	0.002	[-0.001, 0.005]	0.20

* Per 100,000 samples. (Training size was centered at 510,000 as the training data had a mean of 514,293 samples)

** Per 1,000 samples. (Training size for the transfer learned approach was centered at 4,400 as the training data for fine-tuning had a mean of 4,371 samples.)

4.5.2 The Surveillance Error Grid

Surveillance Error Grid analysis showed minimal differences between models and training data compositions. Across all models, racial groups, and training data ratios, almost all predictions fell within the “None” or “Slight” risk zones, while approximately 3–6% were classified as “Moderate” risk and no predictions were classified as “High” or “Extreme” risk (Appendix F). These findings indicate that the small differences in RMSE observed between models were unlikely to be clinically meaningful.

5 Discussion

The overall aim of this thesis was to investigate the potential of CGM data and machine learning methods for cardiovascular risk assessment. To address this aim, the thesis combined a systematic evaluation of the existing literature with the assessment of foundation model-derived glucose representations and an investigation of algorithmic fairness in glucose prediction. The following discussion integrates the findings of these three studies, and considers their methods and results in relation to existing literature.

5.1 Evidence for CGM in Cardiovascular Risk Assessment

Through the literature 53 studies were identified as investigating the link between CGM-data and CVD outcomes. There were very inconsistent findings across the literature. A wide range of CGM-derived metrics were investigated in relation to both clinical and subclinical cardiovascular outcomes, and the statistical adjustment strategies varied considerably between studies. It was observed that studies focusing on people with T2D had bigger study populations and more consistently found evidence for associations compared to studies focusing on people with T1D which may indicate a lack of statistical power in the studies with T1D, which may explain some of the inconsistencies across the studies uncovered in the scoping review. The current evidence base is also limited by its geographical concentration mainly in Asia and Europe and by its almost exclusive focus on association analyses. Only three studies included some prediction done with MAGE which shows a knowledge gap in the literature.

The most frequently investigated CGM-derived metric was TIR, especially in analysis with clinical CVD however inconsistent results were found across studies and similar results were found in³⁶ The second most frequently investigated CGM-derived metric was MAGE. It seemed more consistently associated with coronary artery disease and severity scores, despite the smaller sample size compared to TIR however this is only based on a small section of the included studies. Furthermore, MAGE was the only CGM-derived metric evaluated for its predictive performance, and this was limited to coronary artery disease and subclinical peripheral artery disease. Although MAGE outperformed HbA1c as a predictor in these studies,^{39,Mi2012?} the prediction models demonstrated only modest discriminative ability. This may indicate that within-day glucose variability is a relevant cardiovascular risk factor. However, MAGE is not a complete measure of glucose variability and is biased toward detecting hyperglycemic excursions.^{ScR57?} Consequently, MAGE may underestimate the contribution of hypoglycemia to cardiovascular risk. Whereas some studies have reported associations between hyperglycemia-focused CGM-derived metrics and cardiovascular outcomes,^{65,ScR43?} others have reported associations between hypoglycemia-related CGM-derived metrics and cardiovascular outcomes,^{Magri?,ScR39?,ScR46?,ScR58?,ScR59?,ScR60?} with some concluding that hypoglycemic excursions are associated with macrovascular complications, whereas hyperglycemic excursions are associated with microvascular complications.^{66,Mesa2022?} Together, these findings suggest that although within-day glucose variability may be an important cardiovascular risk factor, both hyperglycemia and hypoglycemia may contribute to cardiovascular risk and should be considered when evaluating CGM-derived metrics.

Another component playing into the heterogeneity of the studies where the difference in how the statistical analysis were performed. CGM-derived metrics were defined calculated over different time periods or split up into ranges and thresholds were changed for $\text{acr}\{\text{TIR}\}$, time below range (TBR) and time above range (TAR) across different studies. Adjustment strategies varied substantially between studies, reflecting differences in study objectives and analytical approaches. Although most studies adjusted for established cardiovascular risk factors, particularly demographic characteristics, medical history, biochemical variables, and anthropometric measures, the covariates varied between studies, limiting direct comparability of the reported associations. Only a minority of studies adjusted for additional CGM-derived metrics. Whether such adjustment is appropriate depends on the research question, as many CGM-derived metrics capture overlapping aspects of glycemic control. Furthermore, no substantial differences were observed between unadjusted and the most extensively adjusted estimates, suggesting that adjustment for additional covariates generally had only a modest influence on the reported associations.

The heterogeneity of the literature, the limited evidence supporting individual CGM-derived metrics, and the challenges of describing glucose variability through CGM-derived metrics provided part of the rationale for evaluating foundation model-derived representations in the subsequent studies, as the complete CGM time series can be used directly without first reducing it to manually engineered CGM-derived metrics. As a result, information on hypoglycemia, hyperglycemia, mean glucose, glucose variability within and between days, and temporal glucose patterns can all be incorporated simultaneously.

5.2 Learning Meaningful Representations from CGM data

Study II showed that CGMformer, the foundation model developed by Lu et al.⁴⁴, captured both long- and short-term glucose fluctuations, as assessed using conventional CGM-derived metrics, in individuals with T1D and T2D. However, the learned representations (embeddings) did not show clear patterns when stratified by CVD-related variables or race. Nevertheless, these findings suggest that CGMformer can generalize beyond its original training population of Chinese individuals with normal glucose tolerance, impaired glucose regulation, and T2D, because it captured clinically meaningful glycemic patterns in both the Danish CANCAN cohort of people with advanced T2D and the US T1DX cohort of people with T1D.

In our study, cosine distances were lower within-participants than between-participants, indicating that the CGM embeddings captured participant-specific patterns. Similar findings have been reported by Lu et al,⁴⁴ who observed higher within-participant cosine similarity, and by Lutsker et al,⁴⁵ who found lower within-participant cosine distances using the GLUFormer foundation model trained on an Israeli cohort from the Human Phenotype Project.⁸⁶ Collectively, these findings support that CGM foundation models learn stable, individualized representations of glucose dynamics while also capturing common glycemic characteristics shared among individuals with similar glucose control.as well as generalize across independent populations and diabetes types. By learning directly from the complete CGM time series, these representations provide an alternative to manually engineered CGM-derived metrics and establish a reusable foundation for downstream clinical applications. Similar transitions from manually extracted metrics to learned representations have driven performance improvements in fields such as computer vision and natural language processing, where pretrained foundation models have become the standard approach for downstream prediction tasks.⁴³ Whether these learned representations translate into improved predictive performance was subsequently investigated in Study II.

5.3 Incremental Value of CGM-derived Information for NT-proBNP Prediction

Although previous studies have successfully used foundation models to predict CVD-related outcomes, including macrovascular complications⁴⁴ and cardiovascular mortality⁴⁵, the learned representations in the present study did not improve prediction of elevated NT-proBNP beyond established clinical predictors and were associated with increased model complexity. The optimism-corrected performance estimates indicate that the more complex models were susceptible to overfitting, reinforcing the conclusion that the learned representations did not provide robust improvements beyond established clinical predictors. This does not necessarily indicate that foundation model-derived representations are unsuitable for cardiovascular prediction. Rather, their performance may depend on the downstream modeling strategy. In the present study, the embeddings were incorporated as fixed predictors in logistic regression models, whereas Lu et al.⁴⁴ fine-tuned (transfer learning like transfer 1 in Figure 1.2) the CGMformer and achieved an AUROC above 0.70 for predicting macrovascular complications. Similarly, Lutsker et al.⁴⁵ demonstrated that a GLUFormer-derived risk score outperformed HbA1c in stratifying individuals according to cardiovascular-related mortality risk. However, the limited sample size and the lack of datasets containing both CGM data and cardiovascular outcomes precluded the use of more complex downstream models and task-specific fine-tuning as performed in Lu et al.,⁴⁴ and Lutsker et al.,⁴⁵ Furthermore, reducing the embeddings to UMAP dimensions may have resulted in the loss of predictive information, which could partly explain the limited performance of the prediction models. The different findings may also reflect differences in the predicted outcomes. Whereas previous studies predicted clinical cardiovascular events or mortality, the present study focused on elevated NT-proBNP, a biomarker that is affected not only by cardiac dysfunction but also by factors such as age, sex and renal function.⁸⁷ Consequently, glucose-derived embeddings may have a weaker relationship with NT-proBNP than with overt cardiovascular outcomes although this remains speculative. Together, these findings suggest that the successful application of foundation model representations to complex clinical outcomes may depend on the downstream modelling strategy and having access to substantially larger datasets.

While learned representations did not translate into improved prediction of elevated NT-proBNP, TIR and MAGE as standalone predictors did. Traditional CGM-derived metrics remain highly relevant for clinical prediction. Metrics such as TIR quantify well-established aspects of glycemic control, are readily understood by clinicians, and are already incorporated into routine diabetes management, particularly TIR. Their simplicity and established clinical interpretation make them useful reference measures when assessing whether more complex learned representations provide additional predictive information. Previous studies have likewise demonstrated prognostic value for MAGE in individuals with acute coronary syndrome^{88,89}, and greater glucose variability has been associated with higher NT-proBNP concentrations in this population⁸⁹. Participants with elevated NT-proBNP in the CANCAN cohort exhibited characteristics consistent with more advanced disease but also had lower BMI and more favorable glycemic profiles. Although these findings may appear counterintuitive, lower BMI has previously been associated with poorer outcomes in patients with established heart failure, while obesity is associated with lower circulating NT-proBNP concentrations⁹⁰. Overall, these findings illustrate that the relationship between glycemic control, body composition, and cardiovascular risk is complex underscoring the importance of assessing CGM-derived measures within multivariable prediction models.

In Study II, when adding TIR and MAGE to the clinical prediction model resulted in a modest improvement in AUROC compared with the model based on clinical variables alone. However, the performance of these models should also be considered relative to a simpler benchmark model such as based on age alone. The simpler baseline model showed substantial predictive ability, consistent with the well-established association between increasing age and NT-proBNP concentrations in both healthy populations and individuals with T2D.⁹¹ Together, these findings suggest that although readily available clinical variables capture much of the information relevant for predicting

elevated NT-proBNP, CGM-data may still provide information that could contribute to future cardiovascular risk assessment.

5.4 Fairness of CGM-based Machine Learning Models

While Studies II and III focused on whether foundation model-derived representations and conventional CGM-derived metrics improved cardiovascular risk prediction, the final aim of the thesis considered another important aspect of machine learning models: whether their performance is equitable across population subgroups.

Although the primary focus of this thesis is cardiovascular risk assessment, glucose forecasting provides an appropriate setting for investigating algorithmic fairness in CGM-based machine learning. In contrast to participant-level cardiovascular prediction, glucose forecasting is trained on a continuous stream of CGM measurements, allowing each participant to contribute numerous prediction windows (samples) over the recording period rather than a single outcome. This results in substantially more training examples, enabling the development and evaluation of more complex machine learning models. Consequently, glucose forecasting provides a suitable framework for studying how training data composition influences model performance across demographic groups. Furthermore, automated insulin delivery (AID) systems are becoming increasingly sophisticated, with the long-term goal of approximating the physiological regulation of glucose by a healthy pancreas. However, the delayed onset and peak action of subcutaneously administered insulin limit the ability of these systems to respond rapidly to changing glucose concentrations.⁹² Consequently, accurate short-term glucose forecasting has become increasingly important, particularly for anticipating postprandial glucose excursions and enabling earlier insulin adjustments to reduce the risk of both hyperglycemia and hypoglycemia.⁹² As prediction models become increasingly integrated into diabetes management, ensuring that they perform equitably across patient populations is essential.⁵¹

Study III demonstrated that varying the racial composition of the training dataset was associated with statistically significant differences in model performance across racial groups when predicting future blood glucose values. Although these differences were statistically significant, they were small and unlikely to result in clinically meaningful disparities. This interpretation was supported by the Surveillance Error Grid, which indicated that the observed differences in predictive performance between racial groups had limited clinical impact. Nevertheless, the findings suggest that training data composition should be considered when developing and evaluating machine learning models intended for clinical use. The observed performance differences appeared to disappear following transfer learning, further suggesting that including data from underrepresented groups may help reduce such differences.

Previous studies have demonstrated the potential of transfer learning to improve machine learning performance across a range of healthcare applications. For example, transfer learning has been associated with reduced performance disparities in multi-ethnic cancer omics datasets, suggesting that it may help mitigate the effects of imbalanced training data.^{Gao?} Similar benefits have been reported for personalized glucose prediction, where transfer learning has improved model performance across several prediction tasks.^{Kushner?,Seo2021?} In contrast, Mohebbi et al,^{Mohebbi?} did not observe an improvement with transfer learning and speculated that the limited size of the fine-tuning dataset may have contributed to this finding. Overall, the present findings are largely consistent with the existing literature, as the transfer learned models achieved the best overall performance while also reducing the performance differences observed between racial groups.

The Surveillance Error Grid results were also comparable to those reported by van Doorn et al,³⁵ who evaluated glucose prediction models in people with type 1 diabetes using a similar experimental setting. This agreement suggests that the clinical risk associated with prediction

errors observed in the present study is consistent with previous evaluations of deep learning-based glucose forecasting models.

5.5 Methodological Considerations

The principal strength and value of the scoping review lie in its comprehensive mapping of the literature, including the identification of more relevant studies than any previous review on the topic, to the best of my knowledge, as well as its differentiation between association and prediction studies, identification of methodological heterogeneity, and characterization of important evidence gaps. However, the heterogeneity identified through this review also limited the feasibility of synthesizing effect estimates consistently across studies. The findings were therefore presented descriptively by mapping the number of studies investigating different CGM-derived metrics and cardiovascular outcomes. Although this approach provides an overview of the scope and distribution of the existing evidence, it does not allow definitive conclusions to be drawn about the strength or consistency of associations between specific CGM-derived metrics and cardiovascular outcomes.

Furthermore, the predominance of cross-sectional studies found in the scoping review further limited the clinical interpretation of the evidence identified. Without temporal separation between CGM exposure and cardiovascular outcomes, it is unclear whether the observed glucose patterns preceded cardiovascular disease or resulted from the disease itself, associated behavioural changes, or intensified treatment. Reverse causality could therefore either produce apparent associations that emerged only after the development of CVD or attenuate pre-existing associations through subsequent treatment and behavioural changes. Because few longitudinal studies were identified, it was not possible to determine whether this systematically influenced the findings or contributed to differences between cross-sectional and longitudinal studies.

Furthermore, the focus in the review was to ensure to include all relevant studies in what was deemed the two most relevant biomedical databases by having a broad search strategy, this might have led to missing relevant studies in other databases.

5.5.1 The CGMformer and Cosine Distances

By externally evaluating CGMformer, developed by Lu et al.,⁴⁴ Study II contributes to the validation of existing models, as recommended by Efthimiou et al.⁹³ and provides evidence regarding the foundation model's generalizability. For the evaluation of the foundation model, by combining comparisons with established CGM-derived metrics, cosine-distance analyses, and visualizations, the study assessed whether the representations captured clinically interpretable glycemic patterns as well as participant-specific characteristics. This provided greater insight into the content and potential utility of the learned representations. A limitation of the cosine-distance analyses was the complex dependence structure of the observations. Participants contributed multiple days of CGM data and could therefore be represented in multiple distance comparisons. Consequently, the observations were not independent, making a conventional Mann–Whitney U test inappropriate. Statistical significance would also be difficult to interpret because the large number of pairwise comparisons does not represent an equally large number of independent observations. Moreover, treating all pairwise distances as independent would overstate the effective sample size and could produce misleadingly small p-values. The cosine-distance results were therefore evaluated descriptively and visually, enabling exploration of potential group differences but not formal statistical inference.

5.5.2 Prediction of NT-proBNP

The use of bootstrapping with optimism correction allowed for robust internal validation, thereby reducing the risk of overfitting and overestimating the results which are important factors to take into consideration when working with real-world limited data. The relatively small sample size restricted the analyses because the available data were deemed insufficient for task-specific fine-tuning of CGMformer, and the embeddings were instead incorporated as fixed predictors in comparatively simple downstream models. Reducing the embeddings to UMAP dimensions made the number of predictors manageable relative to the sample size but may have discarded information relevant to NT-proBNP. Consequently, the available data did not permit evaluation of the embeddings under task-specific fine-tuning, and their potential prognostic value therefore remains undetermined. Furthermore, dichotomizing NT-proBNP at 125 pg/mL may be an imperfect replacement for cardiovascular risk and heart failure, limiting the extent to which the analysis could assess the relationship between CGM-derived representations and clinical cardiovascular outcomes. A more definitive evaluation would require substantially larger longitudinal datasets containing temporally aligned CGM measurements, relevant clinical covariates, and subsequent cardiovascular outcomes. However, such datasets remain rare, limiting opportunities for task-specific fine-tuning or the development of more complex prediction models. Access to these data would enable assessment of whether the embeddings provide prognostic information beyond established cardiovascular risk factors and conventional CGM-derived metrics.

5.5.3 Fairness in Clinical Prediction Models

A key strength of study III was its systematic evaluation of how the racial composition of training data influences predictive performance across groups. By examining multiple training strategies and varying group representation in the training data, the study provides a framework that could also be applied to fairness evaluations involving other populations, outcomes, and data modalities. The use of openly available data, together with publicly shared analytical code, further enhances the transparency and reproducibility of the study and facilitates extension of the approach in future research. However, the study included only non-Hispanic White and Black participants from the United States. Future studies should evaluate potential disparities across additional racial and ethnic groups and in other geographical settings. Fairness evaluations could also consider other demographic characteristics, such as age, given that glycemic control varies across the lifespan.⁹⁴ Although algorithmic fairness is most commonly evaluated for binary classification using established criteria such as demographic parity and equalized odds,⁹⁵ comparable approaches for forecasting tasks remain less developed. Study III therefore provides a novel framework for evaluating disparities in glucose forecasting. However, the analysis was computationally intensive because models were repeatedly trained across multiple racial compositions and training strategies. Even though only a shallow neural network was evaluated, the computational requirements were substantial, which may limit the feasibility of applying this framework to more complex models without access to considerable computing resources. The findings may also depend on the selected 60-minute prediction horizon. Although this horizon was chosen to allow time for potential preventive action, shorter forecasting horizons generally yield greater predictive accuracy. Future studies should determine whether the observed effects of racial composition and transfer learning are consistent across prediction horizons and when using CGM data with higher temporal resolution. In addition, the framework was evaluated using only a single US dataset and was not replicated in an independent cohort. Because the forecasting models were trained separately for each participant, external validation of the fitted models themselves was not directly applicable. However, evaluation in an independent dataset would be necessary to determine whether the observed group-level disparities and the effects of training-data composition generalize to other populations, CGM devices, and healthcare settings.

Conclusion 6

This thesis explored the potential of CGM data and machine learning methods to improve cardiovascular risk assessment. Overall, the findings demonstrate both the opportunities and current limitations of using CGM data for this purpose. Although previous studies have reported associations between CGM-derived metrics and cardiovascular outcomes, the additional predictive value of traditional CGM-derived metrics and foundation model-derived representations beyond established clinical predictors was limited in the present work. The thesis further showed that foundation model-derived representations capture meaningful glycemic patterns and highlighted the importance of evaluating machine learning models across demographic groups. Together, these findings suggest that CGM may have a future role in cardiovascular risk prediction, but larger prospective studies are needed to determine whether both traditionally CGM-derived metrics and more complex methods to extract information from CGM data can meaningfully improve cardiovascular risk stratification in clinical practice.

These conclusions are supported by the findings corresponding to the four specific aims.

Aim 1: The scoping review identified a heterogeneous evidence base characterized by numerous studies of associations between glycemic control and cardiovascular outcomes but few studies evaluating the predictive value of CGM-derived metrics.

Aim 2: The external evaluation of CGMformer showed that its representations captured meaningful glycemic and participant-specific patterns in individuals with T1D and T2D, providing evidence of generalizability beyond the population in which the model was developed on.

Aim 3: Traditional CGM-derived metrics and learned representations from the foundation model provided limited additional value for identifying elevated NT-proBNP beyond established clinical predictors in individuals with T2D.

Aim 4: Glucose-forecasting performance differed between non-Hispanic Black and non-Hispanic White participants. These disparities were influenced by the racial composition of the training data and the modelling strategy, while transfer learning showed potential as an approach for reducing some performance differences.

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Study I: Search strings A

Table A.1: Search in Ovid MEDLINE (Search date: 22 January 2024)

#	Searches	Results
1	Blood Glucose	10241
2	Self-Monitoring/ Continuous Glucose Monitoring/	34
3	((blood-glucose or bloodglucose or glucose) adj2 (monitor\$ or selfmonitor)) <i>or</i> cgmorb _g moriscgmorr _t cgmor(<i>time</i> adj3 <i>range</i>)). <i>ti, ab, kf, kw</i> or bloodsugar\$ or glucose\$ or glyc?emic\$ or sugar). <i>ti, ab, kf, kw.</i>)adj5(<i>flash</i> or <i>fluctuat</i> or intermittent\$ or scan\$ or variabilit). <i>ti, ab, kf, kw.</i> 8669 5 <i>or</i> /1— 4 38366 6 <i>expCardiovascularDiseases/</i> 2760764 7 (cardiovasc or cvd or cardiometabol\$ or cardio-metabol\$ or heart\$ or macrovasc\$ or vasc). <i>ti, ab, kw, kf.</i> 2182473 8 <i>or</i> /6— 7 3971342 9 <i>expDiabetesMellitus/</i> 518864 10 (diabet or dm1 or dm2 or dmi or dmii or iddm or niddm or t1d or t2d). <i>ti,ab,kw,kf.</i> or/9-10	815092
11	or/9-10	875527
12	and/5,8,11	2416
13	exp Animals/ not Humans/	5189493
14	12 not 13 Updated search: 11 March 2025	2300
15	limit 14 to (da="20231201-20261231" or dt="20231201-20261231" or ez="20231201-20261231" or ed="20231201-20261231")	310
16	or/14-15	2610

Table A.2: Search in Embase (Ovid) (Search date: 22 January 2024)

#	Searches	Results
1	blood glucose monitoring/	37574
2	exp continuous glucose monitoring system/	5648
3	((blood-glucose or bloodglucose or glucose) adj2 (monitor\$ or selfmonitor))orcgmorbgmorisecgmorrtecmor(timeadj3range)).ti, ab, kf, kw. 4850 or bloodsugar\$ or glucose\$ or glyc?emic\$ or sugar).ti, ab, kf, kw.)adj5(flashorfluctuat or intermittent\$ or scan\$ or variabilit).ti, ab, kf, kw. 16430 5 or/1–4 81510 6 expcardiovascularisease/ 5122163 7 (cardiovasc or cvd or cardiometabol\$ or cardio-metabol\$ or heart\$ or macrovasc\$ or vasc).ti, ab, kw, kf. 3044925 8 or/6–7 6303633 9 expdiabetesmellitus/ 1239184 10 (diabet or dm1 or dm2 or dmi or dmii or iddm or niddm or t1d or t2d).ti,ab,kw,kf.	1237019
11	or/9-10	1501812
12	and/5,8,11	11217
13	(rat or rats or mouse or mice or swine or porcine or murine or sheep or lambs or pigs or piglets or rabbit or rabbits or cat or cats or dog or dogs or cattle or bovine or monkey or monkeys or trout or marmoset\$1).ti. and animal experiment/	1237966
14	animal experiment/ not (human experiment/ or human/)	2601109
15	or/13-14	2672240
16	12 not 15	10861
17	limit 16 to exclude medline journals Updated search: 11 March 2025	2176
18	limit 17 to (dc="20231201-20261231" or dd="20231201-20261231" or rd="20231201-20261231")	467
19	or/17-18	2643

Study I: Included and excluded reports **B**

C Study I: Geographical location

Asia (n=27) • China (n=19) • Japan (n=8) Europe (n=22) • Italy (n=7) • 3 Belgium (n=3) • 2 Ukraine (n=2) • 2 Finland (n=2) • 2 Spain (n=2) • 1 Greece (n=1) • 1 Malta (n=1) • 1 Sweden (n=1) • 1 The Netherlands (n=1) • 1 Portugal (n=1) • 1 Russia (n=1) North America (n=2) • 2 USA (n=2) Australia / Oceania (n=1) • 1 Australia (n=1)

Borg 2011(ADAG study) had study centers in: • Europe: Netherlands, Denmark, Italy • Africa: 2011 • North America: USA

Borg R, Kuenen J C, Carstensen B, Zheng H, Nathan D M, Heine R J, et al. HbA1(c) and mean blood glucose show stronger associations with cardiovascular disease risk factors than do postprandial glycaemia or glucose variability in persons with diabetes: the A1C-Derived Average Glucose (ADAG) study. 2011; ADAG Study Group, editor. Diabetologia. 54: 69–72. doi:10.1007/s00125-010-1918-2

Study I: Subclinical results



- ☐ Add table with subclinical study design overviews
- ☐ Add table overview of subclinical results
- ☐ Add adjustment column to Table D.1 the same way as I did with the first two studies
- ☐ Add subclinical table from the suppl. of the published ScR study down in supplementary as well

Clinical cardiovascular outcomes comprised cardiovascular mortality, major adverse cardiovascular events, macrovascular complications, coronary artery disease, stroke, and peripheral artery disease. Most studies were observational and differed substantially in study populations, CGM-derived metrics, outcome definitions, and adjustment strategies. Consequently, direct comparison across studies was challenging. The reported effect estimates from the least and most adjusted analyses are summarized in Table D.1.

Table D.1: Main findings of the included studies on clinical cardiovascular outcomes. WIP: Still need to add suppl. files, and adjustment column

Outcome, reference, and continuous glucose monitoring metrics	Unadjusted or least adjusted findings	<i>P</i> value for the least adjusted findings	Most adjusted findings	<i>P</i> value for the most adjusted findings	Adjustments
Cardiovascular Mortality					
Lu et al ⁶⁰ , 2021					Model 1: Demographic
TIR > 85%	HR 1.00	< 0.001 (trend)	HR 1.00	0.02 (trend)	Model 2: Demographic
TIR 71% – 85%	HR: 1.43 (0.95–2.14)	< 0.001 (trend)	HR: 1.35 (0.90–2.04)	0.02 (trend)	Anthropometric
TIR 51% – 70%	HR: 1.66 (1.12–2.45)	< 0.001 (trend)	HR: 1.47 (0.99–2.19)	0.02 (trend)	Lifestyle
TIR ≥ 50%	HR: 2.15 (1.47–3.13)	< 0.001 (trend)	HR: 1.85 (1.25–2.72)	0.02 (trend)	Medical History
TIR as a continuous variable (each 10% decrease)	HR 1.08 (1.03–1.13)	—	HR 1.05 (1.00–1.11)	—	Physiological Biochemical Medication
Wei et al ^{Wei2019?*} , 2019					Least adjusted: Unadjusted
Hypoglycemic events	HR: 2.03 (1.21–3.41)	—	HR: 2.64 (1.40–4.99)	—	Most adjusted: Demographic Anthropometric Lifestyle Medical History Biochemical Medication
Major Adverse Cardiovascular Events					
He et al ^{He2023?} , 2023					
Blood glucose risk index	HR: 0.97 (0.85–1.10)	0.61	HR: 0.98 (0.85–1.13)	0.75	
Low blood glucose index	HR: 2.37 (1.16–4.83)	0.02	HR: 2.73 (1.21–6.16)	0.02	
High blood glucose index	HR: 0.94 (0.81–1.08)	0.38	HR: 0.94 (0.81–1.09)	0.44	
Average of daily risk range	HR: 1.00 (0.93–1.07)	> 0.99	HR: 1.01 (0.93–1.09)	0.80	

Table D.1: Main findings of the included studies on clinical cardiovascular outcomes. WIP: Still need to add suppl. files, and adjustment column

Outcome, reference, and continuous glucose monitoring metrics	Unadjusted or least adjusted findings	<i>P</i> value for the least adjusted findings	Most adjusted findings	<i>P</i> value for the most adjusted findings	Adjustments
GMI	HR: 0.98 (0.91–1.06)	0.65	HR: 0.99(0.91–1.07)	0.78	
M-value	HR: 0.98 (0.91–1.05)	0.54	HR: 0.98(0.91–1.06)	0.64	
Wei et al ^{Wei2019?*} , 2019					
Hypoglycemic events	HR: 1.501 (1.207–1.866)	—	HR: 1.615 (1.239–2.106)	< 0.001	
Macrovascular Complications					
De Meulemeester et al ^{Meulemeester2024?†,*, 2024}					
TIR	OR: 0.939 (0.829–1.063)	> 0.05	OR: 0.896 (0.738–1.087)	> 0.05	
TITR	OR: 0.901 (0.775–1.047)	> 0.05	OR: 0.933 (0.745–1.169)	> 0.05	
Bezerra et al ^{Bezerra2023?*, 2023}					
TIR	OR: 0.66 (0.46–0.93)	0.02	OR: 0.68 (0.39–1.16)	0.15	
Time below 54 mg/dL	OR: 1.10 (0.88–1.38)	0.39	OR: 0.92 (0.62–1.34)	0.65	
TBR	OR: 0.93 (0.80–1.09)	0.39	OR: 0.77 (0.54–1.11)	0.17	
TAR	OR: 1.04 (1.01–1.08)	0.01	OR: 1.04 (0.99–1.10)	0.08	
Time above 250 mg/dL	OR: 1.04 (1.00–1.08)	0.03	OR: 1.03 (0.97–1.09)	0.29	
CV	OR: 1.03 (0.95–1.11)	0.52	OR: 0.92 (0.81–1.06)	0.25	
GMI	OR: 2.17 (1.14–4.11)	0.02	OR: 2.03 (0.77–5.37)	0.15	
Deng et al ^{Deng2023?*, 2023}					
% CV tertile 1 (Hp1; reference)	OR: 1.000	0.07 (interaction)	OR: 1.000	0.008 (interaction)	
% CV tertile 1 (Hp2-2; reference)	OR: 1.000	0.07 (interaction)	OR: 1.000	0.008 (interaction)	
% CV tertile 2 (Hp1)	OR: 1.483 (0.907–2.423)	0.12	OR: 1.048 (0.528–2.078)	0.89	
% CV tertile 2 (Hp2-2)	OR: 1.399 (0.829–2.358)	0.21	OR: 0.659 (0.296–1.466)	0.31	

Table D.1: Main findings of the included studies on clinical cardiovascular outcomes. WIP: Still need to add suppl. files, and adjustment column

Outcome, reference, and continuous glucose monitoring metrics	Unadjusted or least adjusted findings	<i>P</i> value for the least adjusted findings	Most adjusted findings	<i>P</i> value for the most adjusted findings	Adjustments
% CV tertile 3 (Hp1)	OR: 2.347 (1.393–3.957)	0.001	OR: 2.461 (1.183–5.121)	0.02	
% CV tertile 3 (Hp2-2)	OR: 1.217 (0.731–2.027)	0.45	OR: 0.540 (0.245–1.191)	0.13	
El Malahi et al ^{Malahi2022?} , 2022					
TIR	—	—	—	> 0.05	
SD	—	—	—	> 0.05	
CV	—	—	—	> 0.05	
Magri et al ^{66†} , 2018					
TBR	—	—	OR: 1.12 (1.014–1.228)	0.02	
Lowest BG value	—	—	—	—	
Area under the TBR curve	—	—	—	—	
Coronary Artery Disease					
De Meulemeester et al ^{Meulemeester2024?†,*} , 2024					
TITR	OR: 1.039 (0.812–1.330)	> 0.05	OR: 1.255 (0.874–1.803)	> 0.05	
TIR	OR: 1.072 (0.866–1.328)	> 0.05	OR: 1.164 (0.844–1.607)	> 0.05	
Sheng et al ^{Sheng2023?} , 2023					
TIR < 20%	—	—	OR: 2.143 (1.554–3.287)	—	
TIR 20% – 40%	—	—	OR: 1.049 (0.945–2.022)	—	
TIR 40% – 60%	—	—	OR 0.854 (0.495–1.473)	—	

Table D.1: Main findings of the included studies on clinical cardiovascular outcomes. WIP: Still need to add suppl. files, and adjustment column

Outcome, reference, and continuous glucose monitoring metrics	Unadjusted or least adjusted findings	<i>P</i> value for the least adjusted findings	Most adjusted findings	<i>P</i> value for the most adjusted findings	Adjustments
TIR 60% – 80%	—	—	OR: 0.617 (0.423–1.312)	—	
TIR > 80%	—	—	OR: 0.470 (0.143–1.545)	—	
Chen et al ^{Chen2020?†,*} , 2020					
Controls with SD < 1.40 mmol/L, MAGE < 3.90 mmol/L, LAGE < 4.40 mmol/L, MODD < 0.83 mmol/L versus high BG fluctuations (angina pectoris)	—	—	$\chi^2 = 5.797$	0.02	
Controls with SD < 1.40 mmol/L, MAGE < 3.90 mmol/L, LAGE < 4.40 mmol/L, MODD < 0.83 mmol/L versus high BG fluctuations (angina pectoris)	—	—	$\chi^2 = 7.490$	0.006	
Wei et al ^{Wei2019?*} , 2019					
Hypoglycemic events (myocardial Infarction)	HR: 1.901 (1.067–3.389)	—	HR: 1.549 (0.768–3.124)	0.03	
Hypoglycemic events (unstable angina pectoris)	HR: 1.226 (0.857–1.753)	—	HR: 1.218 (0.794–1.869)	0.30	
Shu-Hua et al ^{Hua2012?e} , 2012					
MAGE level (≥ 3.4 mmol/L)	—	—	OR: 2.286 (1.176–4.446)	0.02	
Su et al ^{39*} , 2011					

Table D.1: Main findings of the included studies on clinical cardiovascular outcomes. WIP: Still need to add suppl. files, and adjustment column

Outcome, reference, and continuous glucose monitoring metrics	Unadjusted or least adjusted findings	<i>P</i> value for the least adjusted findings	Most adjusted findings	<i>P</i> value for the most adjusted findings	Adjustments
MAGE $\geq$ 3.4 mmol/L	—	—	OR: 2.612 (1.423–4.831)	0.002	
MAGE	—	—	AUC: 0.618 (0.555–0.680)	0.001	
Gensini Score					
Chen et al ^{Chen2020?†,*} , 2020					
Controls with SD < 1.40 mmol/L, MAGE < 3.90 mmol/L, LAGE < 4.40 mmol/L, MODD < 0.83 mmol/L	—	—	$t = 6.210$	< 0.001	
Watanabe et al ^{Watanabe2017?*} , 2017					
MAGE	—	—	$r = 0.742$	< 0.001	
Shu-Hua et al ^{Hua2012?*} , 2012					
MAGE	—	—	Unstandardized coefficient $\beta = 4.817$; SE = 1.164; Standardized coefficient $t = 2.984$	0.003	
Su et al ^{39*} , 2011					
MAGE	—	—	Unstandardized $\beta = 7.010$; SE = 1.466; standardized $\beta = 0.237$;	< 0.001	

Table D.1: Main findings of the included studies on clinical cardiovascular outcomes. WIP: Still need to add suppl. files, and adjustment column

Outcome, reference, and continuous glucose monitoring metrics	Unadjusted or least adjusted findings	<i>P</i> value for the least adjusted findings	Most adjusted findings	<i>P</i> value for the most adjusted findings	Adjustments
<i>t</i> = 4.783					
Syntax Score					
Watanabe et al ^{Watanabe2017?*} , 2017					
MAGE	—	—	<i>r</i> = 0.776	< 0.001	
Zhang et al ^{68*} , 2013					
MAGE	—	—	<i>r</i> = 0.518	0.01	
BG fluctuations from 00:00 to 03:00	—	—	<i>r</i> = −0.442	0.04	
BG fluctuations from 03:00 to 06:00	—	—	<i>r</i> = −0.208	0.34	
BG fluctuations from 06:00 to 08:00	—	—	<i>r</i> = 0.678	< 0.001	
BG fluctuations from 08:00 to 11:00	—	—	<i>r</i> = 0.115	0.60	
BG fluctuations from 11:00 to 13:00	—	—	<i>r</i> = 0.523	0.01	
BG fluctuations from 13:00 to 17:00	—	—	<i>r</i> = 0.257	0.24	
BG fluctuations from 17:00 to 19:00	—	—	<i>r</i> = 0.358	0.09	
BG fluctuations from 19:00 to 24:00	—	—	<i>r</i> = −0.018	0.93	
Stroke					
De Meulemeester et al ^{Meulemeester2024?†,*} , 2024					

Table D.1: Main findings of the included studies on clinical cardiovascular outcomes. WIP: Still need to add suppl. files, and adjustment column

Outcome, reference, and continuous glucose monitoring metrics	Unadjusted or least adjusted findings	<i>P</i> value for the least adjusted findings	Most adjusted findings	<i>P</i> value for the most adjusted findings	Adjustments
TITR	OR 0.651 (0.470–0.902)	< 0.05	OR 0.546 (0.347–0.858)	< 0.01	
TIR	OR 0.749 (0.588–0.955)	< 0.05	OR 0.617 (0.440–0.866)	< 0.01	
Guo et al ^{Guo2021?*} , 2021					
TIR: Q1 ($\leq 46\%$; reference)	OR: 1.00	< 0.001	OR: 1.00	< 0.001	
TIR: Q2 (46% – 65%)	OR: 0.86 (0.72–0.95)	< 0.001	OR: 0.80 (0.68–0.92)	< 0.001	
TIR: Q3 (65% – 81%)	OR: 0.71 (0.61–0.81)	< 0.001	OR: 0.64 (0.53–0.79)	< 0.001	
TIR: Q4 ($> 81\%$)	OR: 0.66 (0.58–0.80)	< 0.001	OR: 0.59 (0.50–0.74)	< 0.001	
TIR (per 10% increase)	OR: 0.93 (0.85–0.98)	0.008	OR: 0.89 (0.82–0.95)	< 0.001	
Wei et al ^{Wei2019?*} , 2019					
Hypoglycemic events	HR: 1.691 (1.144–2.499)	—	HR: 1.813 (1.110–2.960)	0.06	
Peripheral artery disease					
De Meulemeester et al ^{Meulemeester2024?†,*} , 2024					
TITR	OR 0.680 (0.426–1.085)	> 0.05	OR 0.807 (0.382–1.703)	> 0.05	
TIR	OR 0.736 (0.520–1.042)	> 0.05	OR 0.811 (0.470–1.398)	> 0.05	
Lower Extremity Arterial Disease					
Li et al ^{Li2020?} , 2020					
TIR	OR: 0.979 (0.968–0.991)	< 0.001	OR: 0.979 (0.965–0.992)	0.002	
CV	OR: 1.040 (1.003–1.078)	0.04	OR: 1.038 (0.996–1.081)	0.08	

Table D.1: Main findings of the included studies on clinical cardiovascular outcomes. WIP: Still need to add suppl. files, and adjustment column

Outcome, reference, and continuous glucose monitoring metrics	Unadjusted or least adjusted findings	<i>P</i> value for the least adjusted findings	Most adjusted findings	<i>P</i> value for the most adjusted findings	Adjustments
SD	OR: 1.325 (1.038–1.691)	0.02	OR: 1.158 (0.824–1.627)	0.40	
TIR-without LEAD (1)	OR: 1.00	—	OR: 1.00	—	
TIR-mild LEAD (1)	OR: 0.98 (0.97–1.00)	0.14	OR: 0.99 (0.97–1.01)	0.25	
TIR-moderate LEAD (1)	OR: 0.97 (0.95–0.99)	0.007	OR: 0.97 (0.95–0.99)	0.01	
TIR-without severe LEAD (1)	OR: 0.96 (0.94–0.98)	0.002	OR: 0.96 (0.94–0.98)	0.003	
CV-without LEAD	—	—	OR: 1.00	—	
CV-mild LEAD	—	—	OR: 1.03 (0.98–1.07)	0.28	
CV-moderate LEAD	—	—	OR: 1.02 (0.96–1.09)	0.48	
CV-without severe LEAD	—	—	OR: 1.02 (0.95–1.09)	0.60	
TIR-without LEAD (2)	—	—	OR: 1.00	—	
TIR-mild LEAD (2)	—	—	OR: 0.97 (0.96–1.08)	0.06	
TIR-moderate LEAD (2)	—	—	OR: 0.98 (0.95–0.99)	0.01	
TIR-without severe LEAD (2)	—	—	OR: 0.97 (0.95–0.99)	0.02	
SD-without LEAD	—	—	OR: 1.00	—	
SD-mild LEAD	—	—	OR: 0.88 (0.47–1.64)	0.69	
SD-moderate LEAD	—	—	OR: 1.28 (0.58–3.07)	0.58	
SD-without severe LEAD	—	—	OR: 1.52 (0.92–2.41)	0.10	

All hazard ratios (HRs), odds ratios (ORs), and areas under the curve (AUCs) are reported as follows: point estimate (95% CI).

* This study appears multiple times as it investigated multiple cardiovascular disease outcomes.

† This study appears in both the clinical and subclinical disease outcome tables owing to the investigation of multiple cardiovascular disease outcomes.

The full table with adjustments is provided in **Multimedia Appendix 10**. Further elaboration on the adjusted variables can be found in **Multimedia Appendices 7 and 8**.

*Abbreviations: * nr: Not applicable or not available/not reported, TIR: time in range, GMI: glucose management indicator, TITR: time in tight range, TBR: time below range, TAR: time above range, CV: coefficient of variation, BG: blood glucose, MAGE: mean amplitude of glycemic excursions, LAGE: largest amplitude of glycemic excursions, MODD: mean of daily differences, LEAD: lower extremity arterial disease.

E Study II: Prediction stability

Prediction stability plots (?@fig-pred__stability) revealed more stability the fewer predictors.

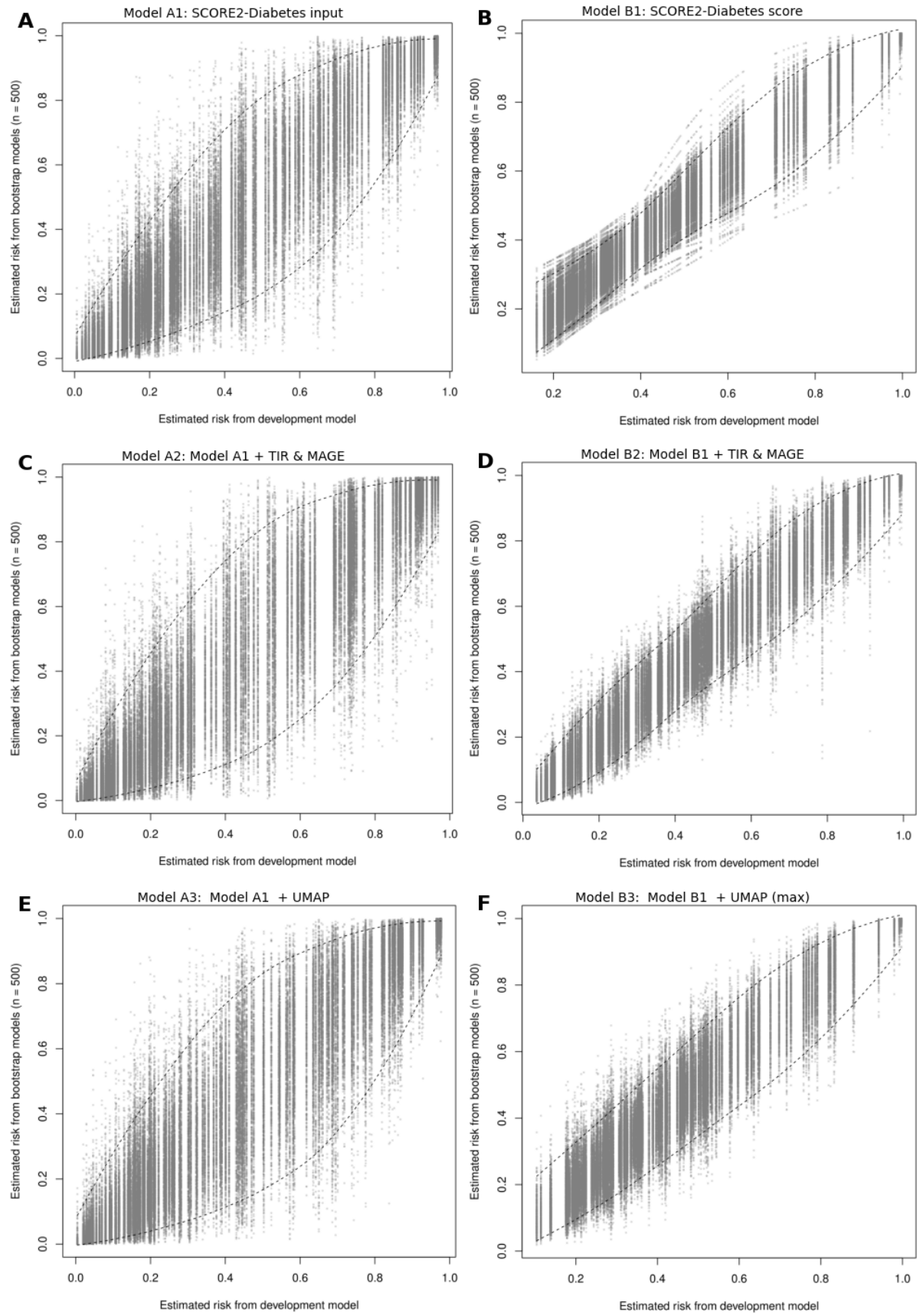


Figure E.1: Prediction stability plots show the predictions (estimated risk) for the full dataset from the development model (equivalent to the apparent model) on the x-axis, and the predictions from all bootstrap models on the y-axis. Tight alignment along the diagonal indicates higher prediction stability.

F Study III: Surveillance Error Grid Results

Table F.1: Results from the Surveillance Error Grid for White participants. Values are reported as percentages.

Group	Model	Zone	0	20	40	60	80	100
White	LOCF	None	69%	69%	69%	69%	69%	69%
		Slight	26%	26%	26%	26%	26%	26%
		Moderate	4%	4%	4%	4%	4%	4%
		High	0%	0%	0%	0%	0%	0%
		Extreme	0%	0%	0%	0%	0%	0%
	Base-Individual	None	66%	66%	66%	66%	66%	66%
		Slight	28%	28%	28%	28%	28%	28%
		Moderate	6%	6%	6%	6%	6%	6%
		High	0%	0%	0%	0%	0%	0%
		Extreme	0%	0%	0%	0%	0%	0%
	Base-Generalized	None	72%	73%	73%	73%	73%	73%
		Slight	25%	25%	25%	24%	24%	24%
		Moderate	3%	3%	3%	3%	3%	3%
		High	0%	0%	0%	0%	0%	0%
		Extreme	0%	0%	0%	0%	0%	0%
	Transfer Learned	None	72%	72%	73%	73%	73%	73%
		Slight	25%	24%	24%	24%	24%	24%
		Moderate	3%	3%	3%	3%	3%	3%
		High	0%	0%	0%	0%	0%	0%
		Extreme	0%	0%	0%	0%	0%	0%

□ maybe add this one to appendix, since I think they show the same things

Table F.2: Results from the Surveillance Error Grid for Black participants. Values are reported as percentages.

Group	Model	Zone	0	20	40	60	80	100
Black	LOCF	None	71%	71%	71%	71%	71%	71%
		Slight	24%	24%	24%	24%	24%	24%
		Moderate	4%	4%	4%	4%	4%	4%
		High	0%	0%	0%	0%	0%	0%
		Extreme	0%	0%	0%	0%	0%	0%
	Base-Individual	None	66%	66%	66%	66%	66%	66%
		Slight	27%	27%	27%	27%	27%	27%
		Moderate	6%	6%	6%	6%	6%	6%
		High	0%	0%	0%	0%	0%	0%
		Extreme	0%	0%	0%	0%	0%	0%

Group	Model	Zone	0	20	40	60	80	100
	Base -Generalized	Extreme	0%	0%	0%	0%	0%	0%
		None	75%	74%	75%	75%	74%	75%
		Slight	23%	23%	22%	22%	22%	22%
		Moderate	3%	3%	3%	3%	3%	3%
		High	0%	0%	0%	0%	0%	0%
	Transfer Learned	Extreme	0%	0%	0%	0%	0%	0%
		None	75%	74%	75%	75%	74%	75%
		Slight	22%	23%	22%	22%	22%	22%
		Moderate	3%	3%	3%	3%	3%	3%
		High	0%	0%	0%	0%	0%	0%
		Extreme	0%	0%	0%	0%	0%	0%

G Full papers

G.1 Paper I

G.2 Paper II

G.3 Paper III